



Signals and Systems EE252, Spring 2026

Instructor: Tariq Mumtaz

Semester week # 12





Course Learning Outcome (CLOs):

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate the response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-3	Analyze DT signals and systems in the transform domain using the Z transform.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms, and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
CLO-5	Investigate complex engineering problems in signals and systems and formally present the results.	PLO-4 (Investigation)	Cognitive Level-5 (Evaluating)



Roadmap till Semester end:

Frequency Domain Analysis of CT/DT Signals & Systems:

Z- Transform (Wk 7 – 11)

→ Laplace Transform (Wk 12 – 13)

Fourier Transform (Wk 14 – 15)

Project presentations (Wk16)



Learning outcome of Wk#12 session:

The time-domain analysis of linear, time-invariant, continuous-time (LTIC) systems.

Time Domain Analysis (Chapter 02)

- CT System modeling by Differential equations
- CT signal and system representations
- CT system impulse response $h(t)$
- CT Convolution- $y(t) = x(t) * h(t)$
- Properties of CT systems using $h(t)$:
 - Causality
 - BIBO Stability

Frequency Domain Analysis (Chapter 04)

- Eigenfunction of CT systems
- The Laplace Transform



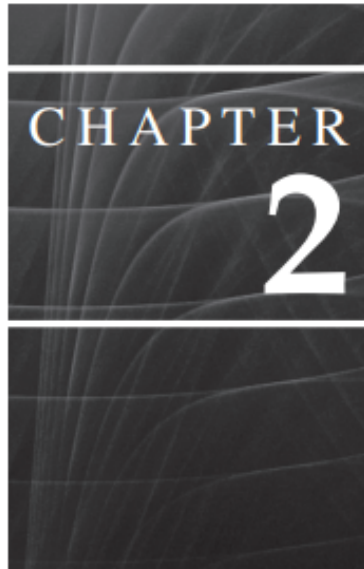
Time Domain analysis- LTIC Systems:

A quick comparison with LTI CT and DT systems:

- Notion of time changes from n to t , i.e., $y[n] \rightarrow y(t)$
- The summation operation $\sum x[n]$ changes to Integration $\int x(t)$
- The difference equations change to Differentiation equations to model the CT system
- Eigenfunction Z^n ($z = \gamma e^{j\Omega}$) changes to e^{st} ($s = \sigma + j\omega$)

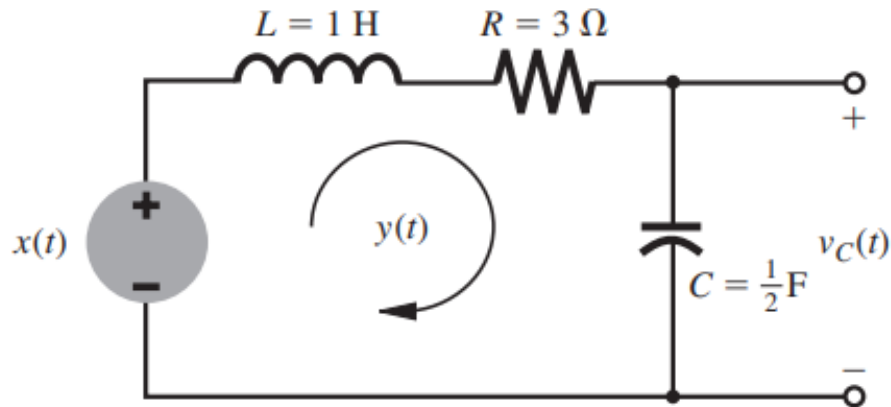


Time Domain analysis- LTIC Systems:



TIME-DOMAIN ANALYSIS OF CONTINUOUS-TIME SYSTEMS

Time Domain analysis- LTIC Systems:



Application of Kirchhoff's voltage law around the loop yields

$$v_L(t) + v_R(t) + v_C(t) = x(t)$$

$$\frac{dy(t)}{dt} + 3y(t) + 2 \int_{-\infty}^t y(\tau) d\tau = x(t)$$

Differentiating both sides of this equation, we obtain

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}$$

This differential equation is the input–output relationship between the output $y(t)$ and the input $x(t)$.

Time Domain analysis- LTIC Systems:

DT vs CT signal representation – Example: Impulse Signal

DT

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Shifting (sampling) property of Impulse

$$x[n]\delta[n - k] = x[k]\delta[n - k]$$

$$\sum_{n=-\infty}^{\infty} x[n]\delta[n - k] = x[k]$$

CT

$$\delta(t) = 0 \quad \text{for } t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\delta(t - \tau)\phi(t) = \delta(t - \tau)\phi(\tau)$$

$$\int_{-\infty}^{\infty} \phi(t)\delta(t - \tau) dt = \phi(\tau)$$



Time Domain analysis- LTIC Systems:

DT vs CT signal representation – Example: Impulse Signal

Convolution Property of Impulse

$$x[n] * \delta[n] = x[n]$$

$$x(t) * \delta(t) = x(t)$$

Arbitrary Signal Representation Using Impulse

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

A discrete time signal can be represented as a sum of scaled and shifted impulses.

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

A continuous-time signal can be represented as an integral of scaled and shifted impulses.

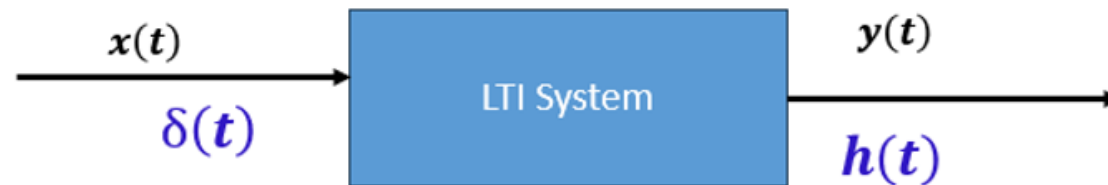


Time Domain analysis- LTIC Systems:

CT LTI System and Impulse Response

Convolution

$$y(t) = x(t) * h(t)$$



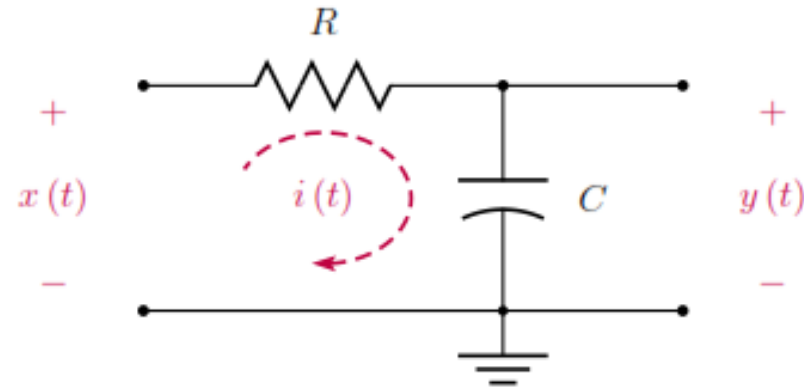
$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

Time Domain analysis- LTIC Systems:

Pulse response of RC circuit revisited

Using convolution, determine the response of the RC circuit to a unit-pulse input signal $x(t) = \Pi(t)$.

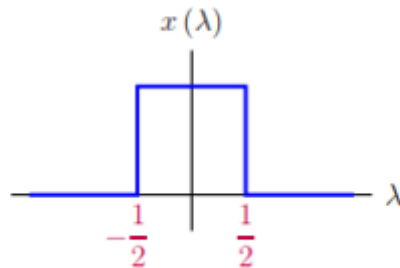


$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

Time Domain analysis- LTIC Systems:

Continuous time convolution process

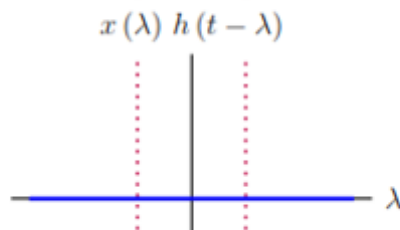
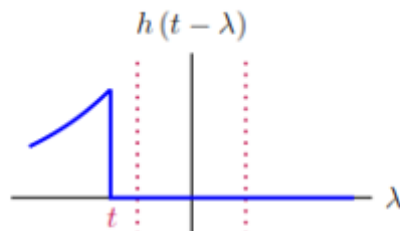
Case 1: $t \leq -\frac{1}{2}$



Case 1: $t \leq -\frac{1}{2}$

Functions $x(\lambda)$ and $h(t - \lambda)$ do not overlap. Therefore

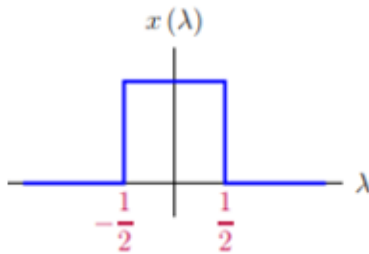
$$y(t) = 0, \quad \text{for } t \leq -\frac{1}{2}$$



Time Domain analysis- LTIC Systems:

Continuous time convolution process

Case 2: $-\frac{1}{2} < t \leq \frac{1}{2}$

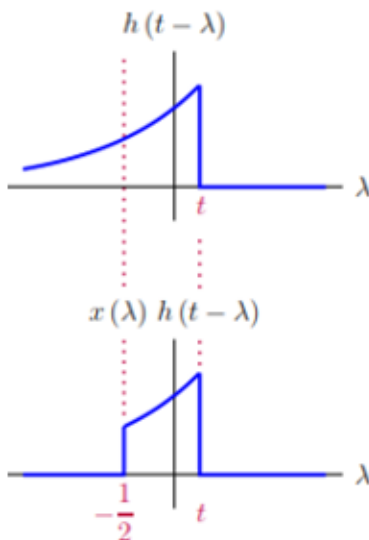


$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

Case 2: $-\frac{1}{2} < t \leq \frac{1}{2}$

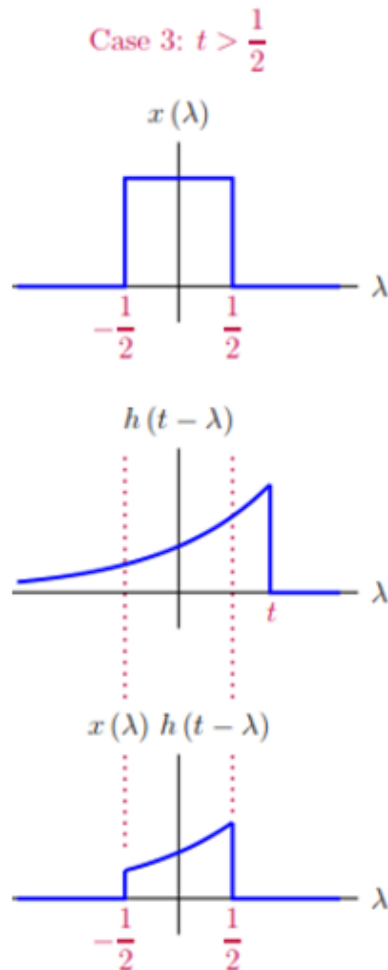
Functions $x(\lambda)$ and $h(t - \lambda)$ overlap in the range $-\frac{1}{2} < \lambda \leq t$. Therefore

$$y(t) = \int_{-1/2}^t \frac{1}{RC} e^{-(t-\lambda)/RC} d\lambda = \left(1 - e^{-(t+1/2)/RC}\right), \quad \text{for } -\frac{1}{2} < t \leq \frac{1}{2}$$



Time Domain analysis- LTIC Systems:

Continuous time convolution process



Case 3: $t > \frac{1}{2}$

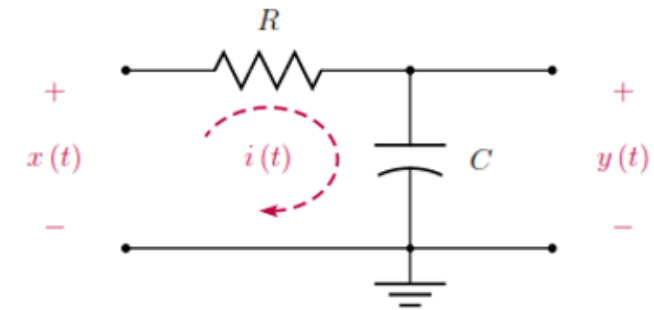
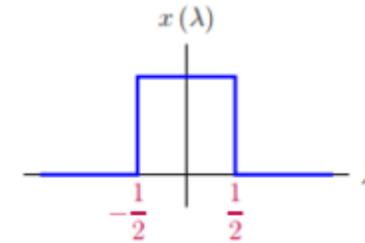
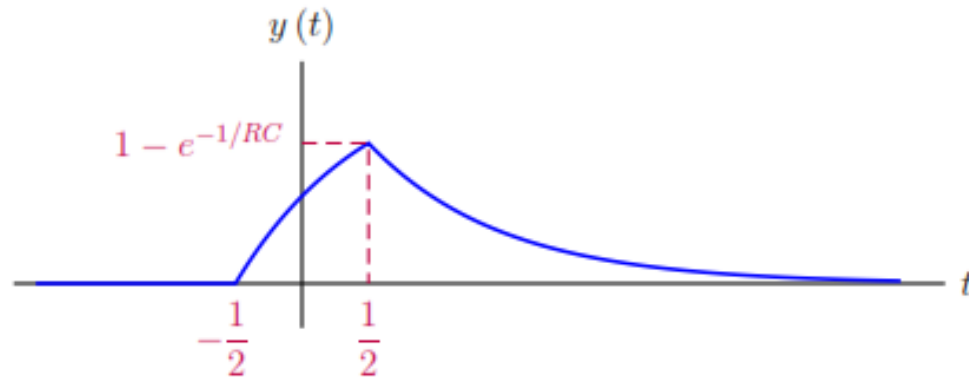
Functions $x(\lambda)$ and $h(t-\lambda)$ overlap in the range $-\frac{1}{2} < \lambda \leq \frac{1}{2}$. Therefore

$$y(t) = \int_{-1/2}^{1/2} \frac{1}{RC} e^{-(t-\lambda)/RC} d\lambda = e^{-t/RC} \left(e^{1/2RC} - e^{-1/2RC} \right), \quad \text{for } t > \frac{1}{2}$$

Time Domain analysis- LTIC Systems:

Continuous time convolution process

$$y(t) = \begin{cases} 0, & t \leq -\frac{1}{2} \\ (1 - e^{-(t+1/2)/RC}), & -\frac{1}{2} < t \leq \frac{1}{2} \\ e^{-t/RC} (e^{1/2RC} - e^{-1/2RC}), & t > \frac{1}{2} \end{cases}$$



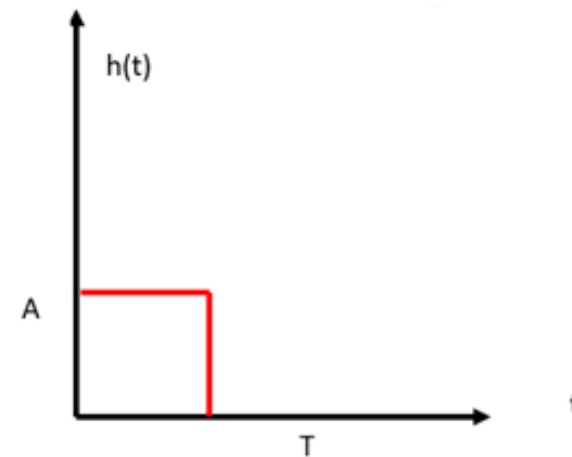
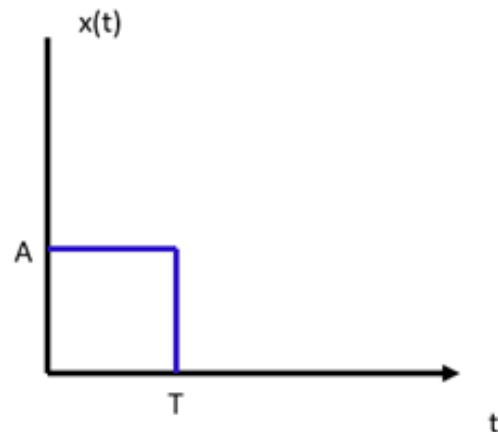
Time Domain analysis- LTIC Systems:

Continuous time convolution process

Consider the convolution of the following two signals:

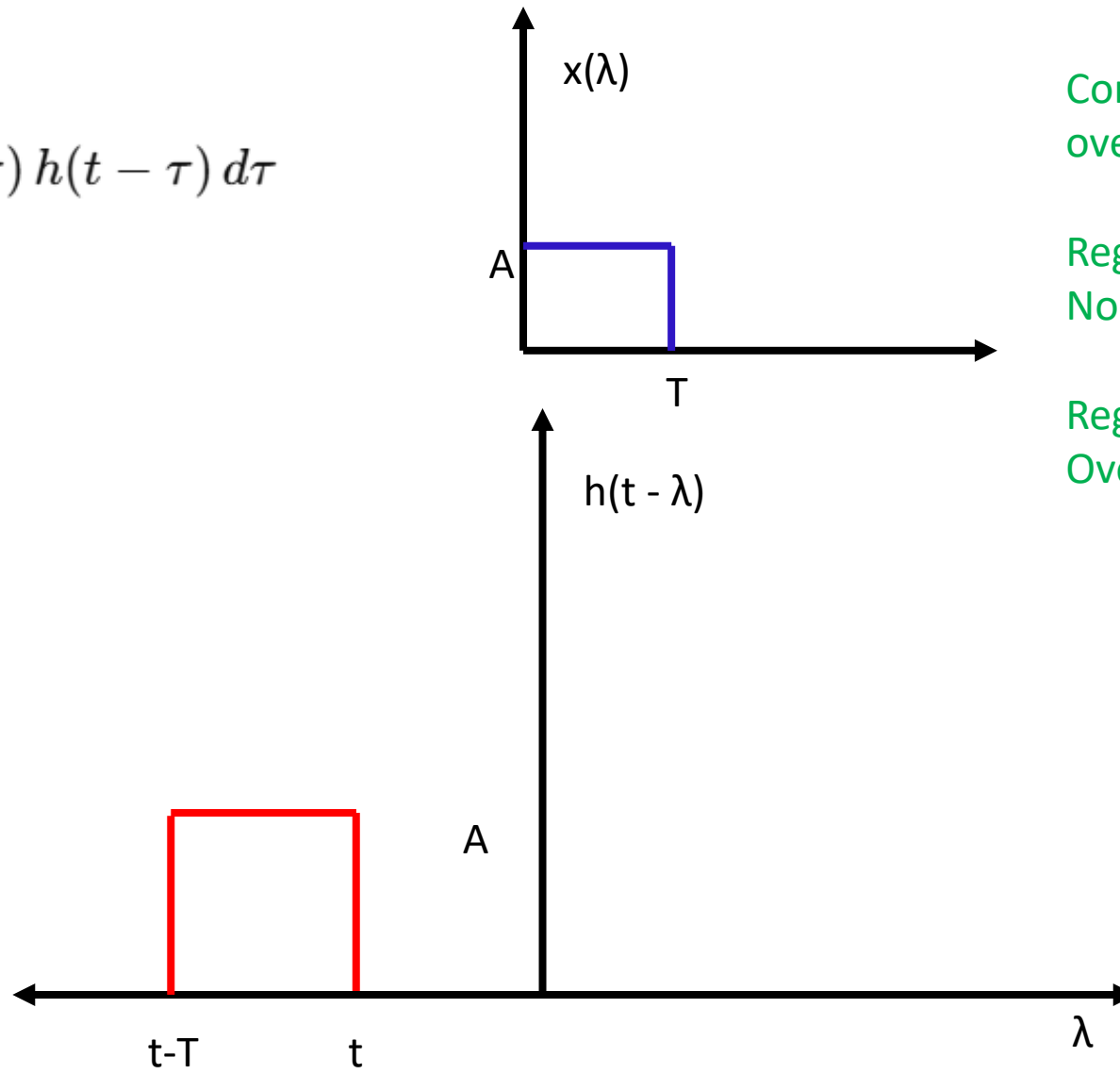
$$x(t) = \begin{cases} A & 0 < t < T \\ 0, & \text{otherwise} \end{cases},$$

$$h(t) = \begin{cases} A & 0 < t < T \\ 0, & \text{otherwise} \end{cases},$$



Time Domain analysis- LTIC Systems:

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$



Come up with non- overlapping and overlapping regions

Region 1: t is less than or equal 0
No- overlapping

Region 2: $0 < t < T$
Overlapping region

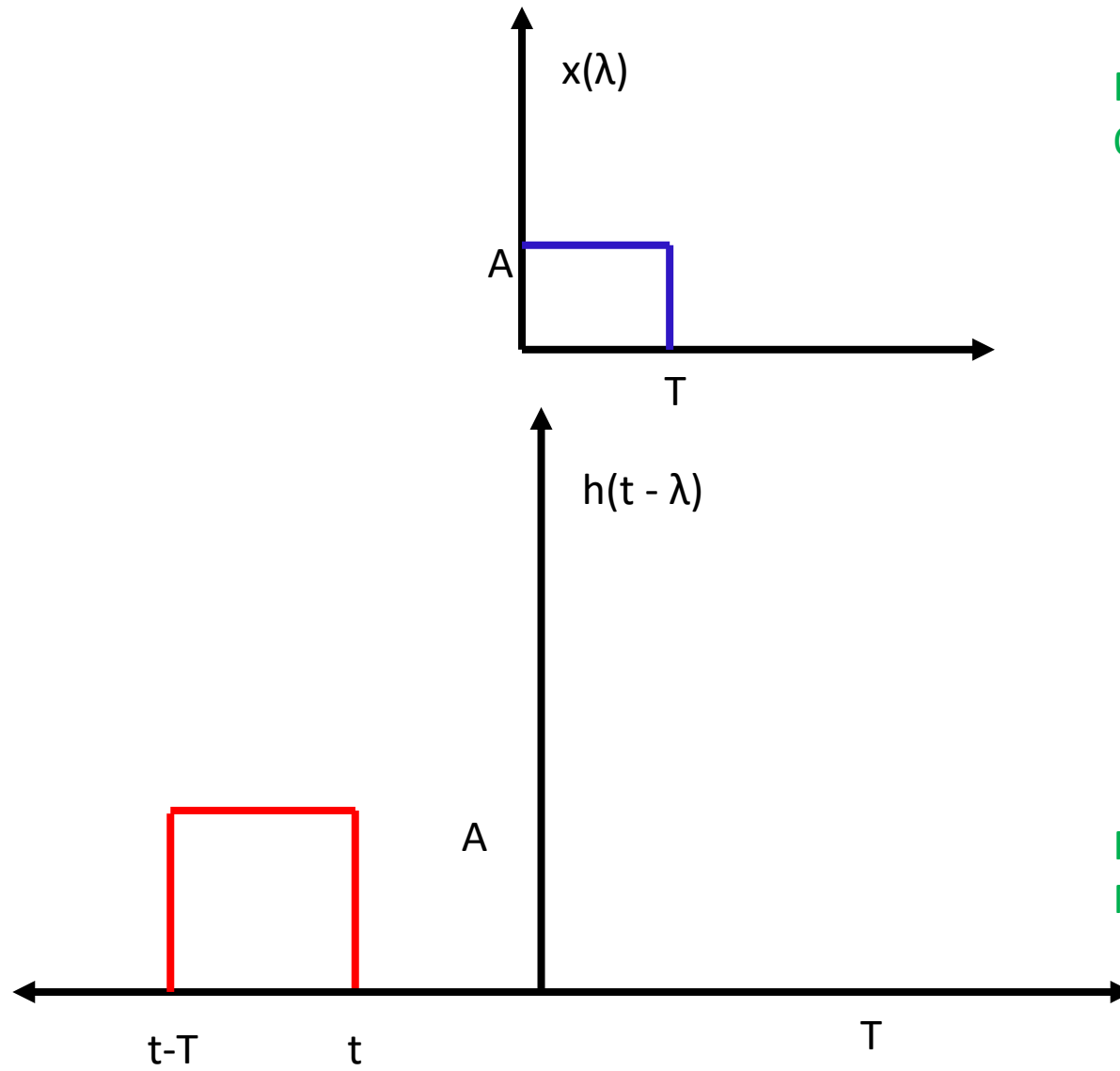
$$y(t) = \int_0^t A \cdot A d\tau$$

$$y(t) = A^2 \int_0^t d\tau$$

$$y(t) = A^2 [\tau]_0^t$$

$$y(t) = A^2 (t - 0) = A^2 t$$

Time Domain analysis- LTIC Systems:



Region 3: $t > T$
Overlapping region

$$y(t) = \int_{t-T}^T A \cdot A d\tau$$

$$y(t) = A^2 \int_{t-T}^T d\tau$$

$$y(t) = A^2 [\tau]_{t-T}^T$$

$$y(t) = A^2 (T - (t - T))$$

$$y(t) = A^2 (2T - t)$$

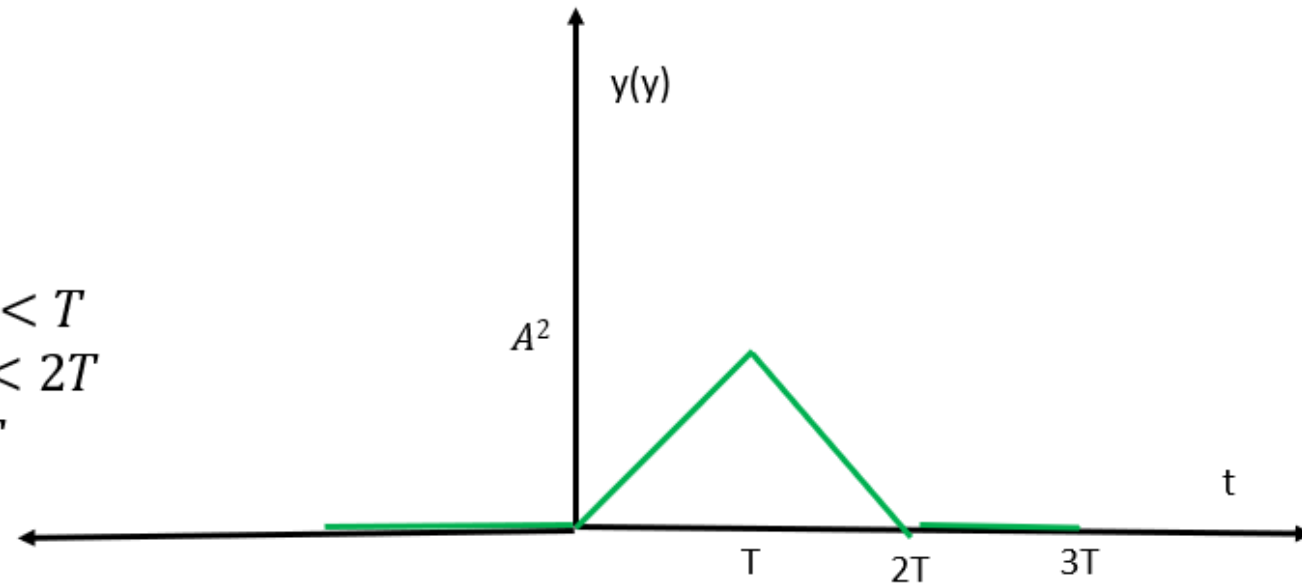
Region 4: $t > 2T$
No- overlapping

Time Domain analysis- LTIC Systems:

Continuous time convolution process

$y(t)$ for all t

$$\begin{aligned} y(t) &= 0 & \text{for } t < 0 \\ &= A^2 t & 0 < t < T \\ &= A^2 (2T - t) & T < t < 2T \\ &= 0 & t > 2T \end{aligned}$$

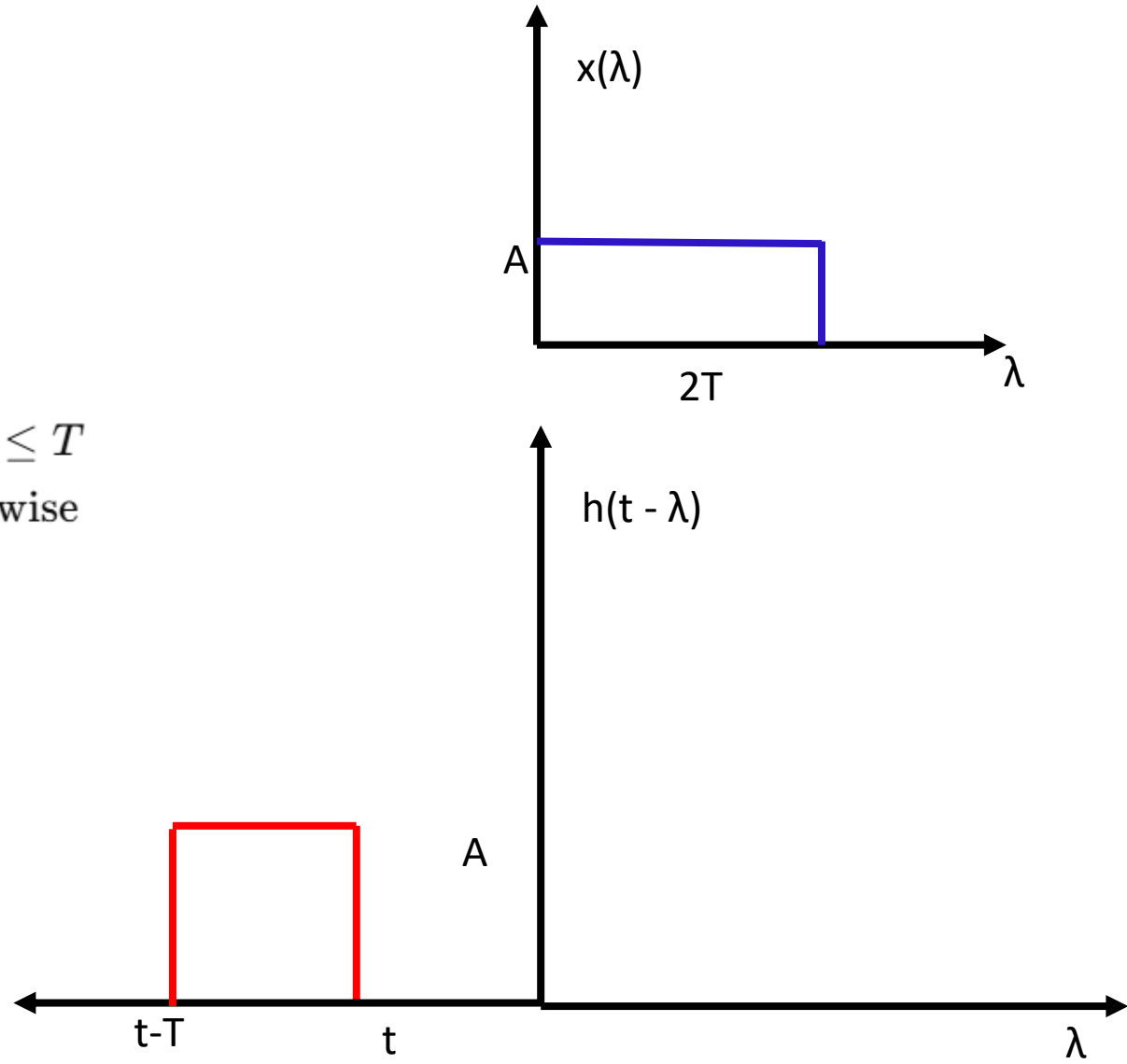


Time Domain analysis- LTIC Systems:

Continuous time convolution process

Practice Problem:

$$x(t) = \begin{cases} A, & 0 \leq t \leq 2T \\ 0, & \text{otherwise} \end{cases} \quad h(t) = \begin{cases} A, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$



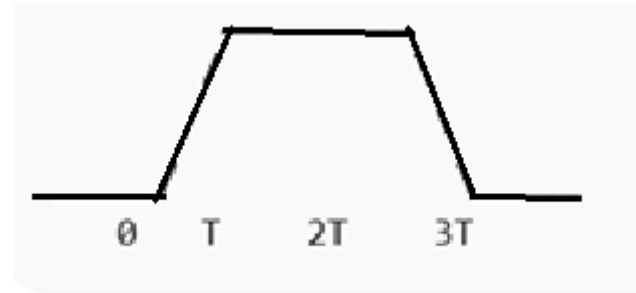
Time Domain analysis- LTIC Systems:

Continuous time convolution process

$$x(t) = \begin{cases} A, & 0 \leq t \leq 2T \\ 0, & \text{otherwise} \end{cases} \quad h(t) = \begin{cases} A, & 0 \leq t \leq T \\ 0, & \text{otherwise} \end{cases}$$

Answer Key: There will be five regions in total.

$$y(t) = \begin{cases} 0, & t < 0 \\ A^2 t, & 0 \leq t \leq T \\ A^2 T, & T \leq t \leq 2T \\ A^2 (3T - t), & 2T \leq t \leq 3T \\ 0, & t > 3T \end{cases}$$



This convolution result will be a trapezoid.



Time Domain analysis- LTIC Systems:

Continuous time convolution process

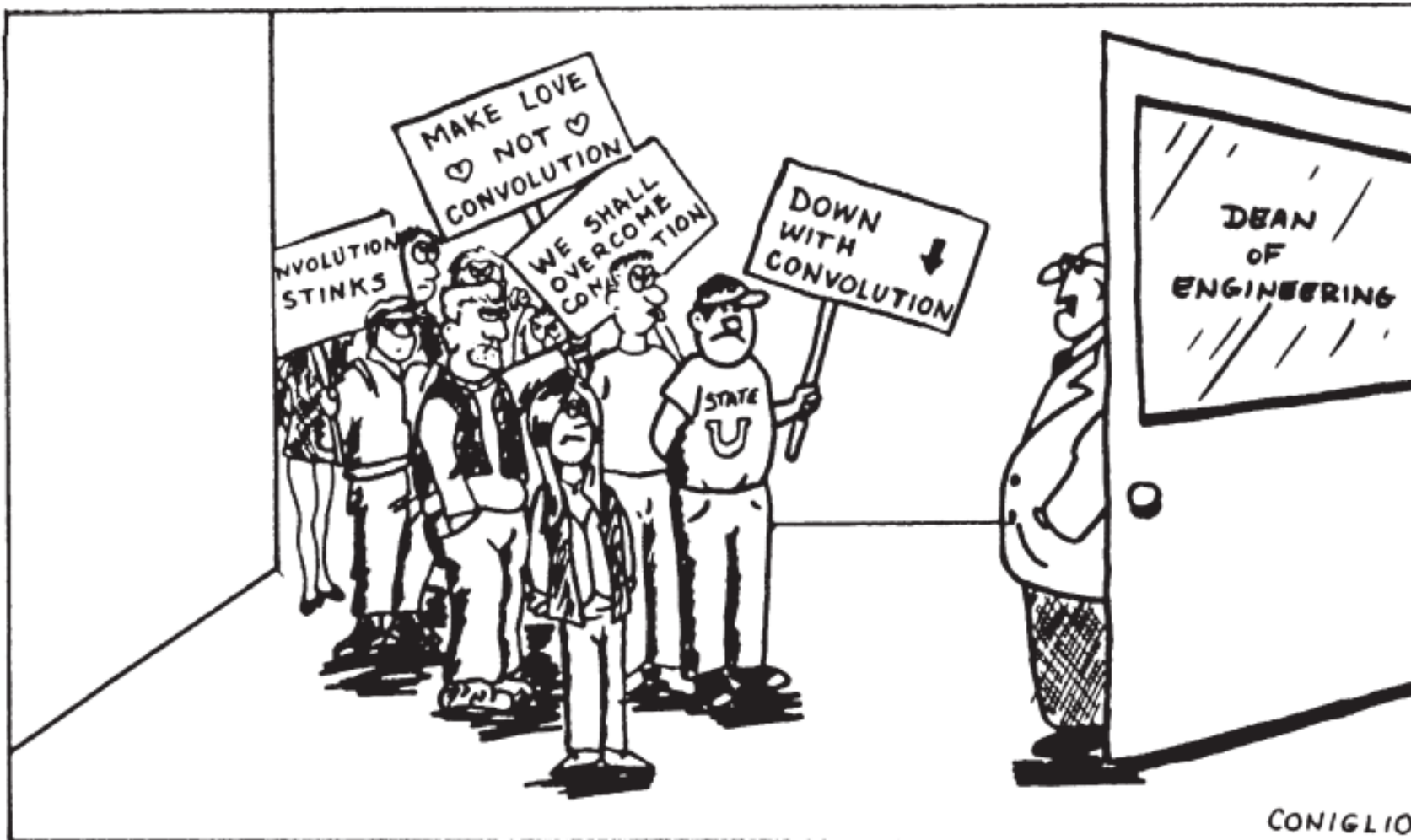
SUMMARY OF THE GRAPHICAL PROCEDURE

The procedure for graphical convolution can be summarized as follows:

1. Keep the function $x(\tau)$ fixed.
2. Visualize the function $g(\tau)$ as a rigid wire frame, and rotate (or invert) this frame about the vertical axis ($\tau = 0$) to obtain $g(-\tau)$.
3. Shift the inverted frame along the τ axis by t seconds. The shifted frame now represents $g(t - \tau)$.
4. The area under the product of $x(\tau)$ and $g(t - \tau)$ (the shifted frame) is $c(t)$, the value of the convolution at t .
5. Repeat this procedure, shifting the frame by different values (positive and negative) to obtain $c(t)$ for all values of t .

Same process as DT convolution – Sliding Tape method

Time Domain analysis- LTIC Systems:



Convolution: Its bark is worse than its bite!



Time Domain analysis- LTIC Systems:

Convolution– Using Table

TABLE 2.1 Select Convolution Integrals

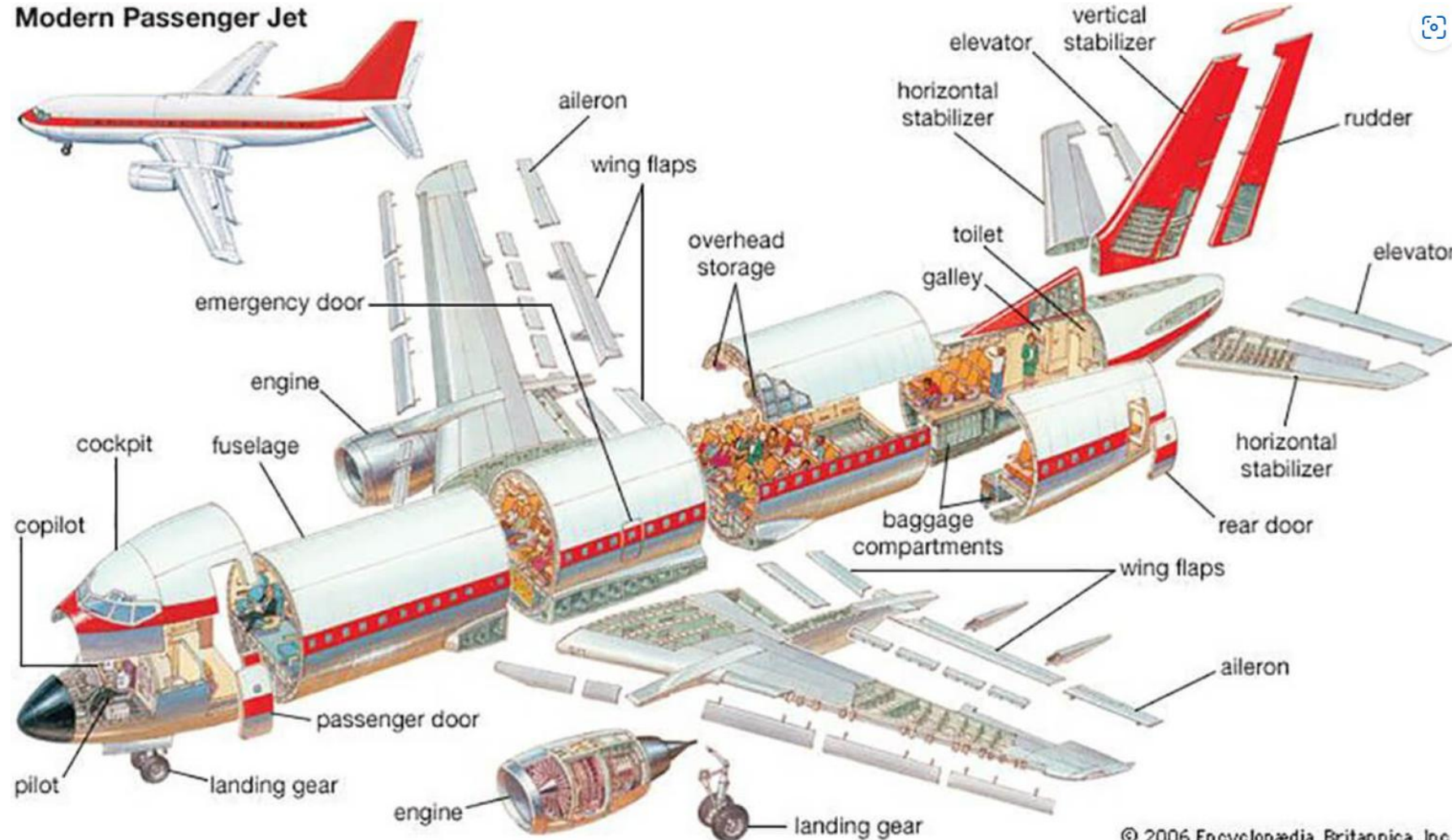
No.	$x_1(t)$	$x_2(t)$	$x_1(t) * x_2(t) = x_2(t) * x_1(t)$
1	$x(t)$	$\delta(t - T)$	$x(t - T)$
2	$e^{\lambda t} u(t)$	$u(t)$	$\frac{1 - e^{\lambda t}}{-\lambda} u(t)$
3	$u(t)$	$u(t)$	$tu(t)$
4	$e^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(t)$	$\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} u(t) \quad \lambda_1 \neq \lambda_2$
5	$e^{\lambda t} u(t)$	$e^{\lambda t} u(t)$	$te^{\lambda t} u(t)$
6	$te^{\lambda t} u(t)$	$e^{\lambda t} u(t)$	$\frac{1}{2} t^2 e^{\lambda t} u(t)$
7	$t^N u(t)$	$e^{\lambda t} u(t)$	$\frac{N! e^{\lambda t}}{\lambda^{N+1}} u(t) - \sum_{k=0}^N \frac{N! t^{N-k}}{\lambda^{k+1} (N-k)!} u(t)$
8	$t^M u(t)$	$t^N u(t)$	$\frac{M! N!}{(M+N+1)!} t^{M+N+1} u(t)$

Complex System Analysis



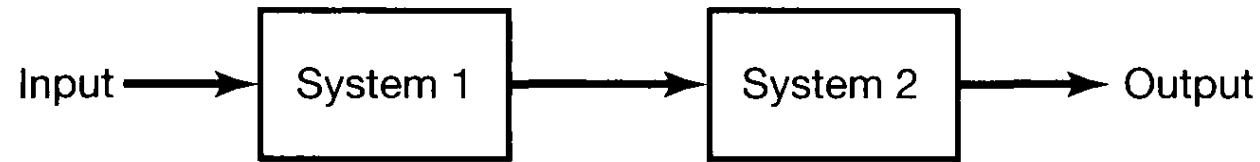
Complex System Analysis

Modern Passenger Jet

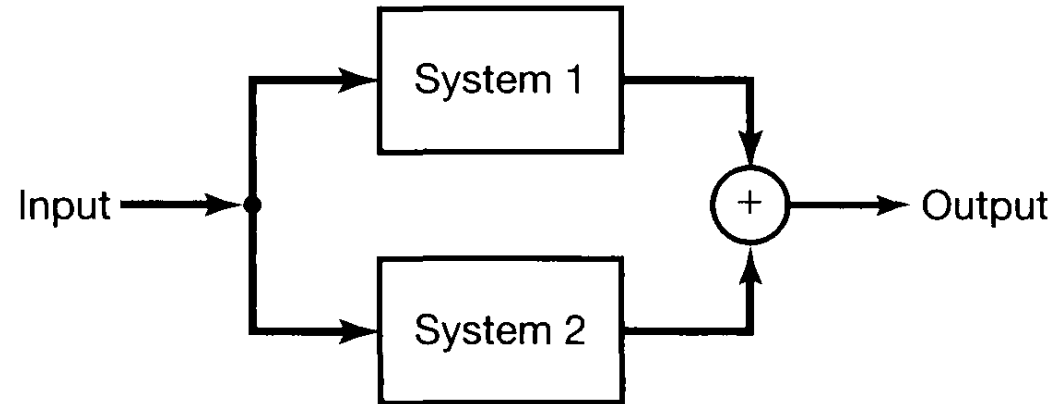


Interconnection of System:

- Series/Cascade Interconnection

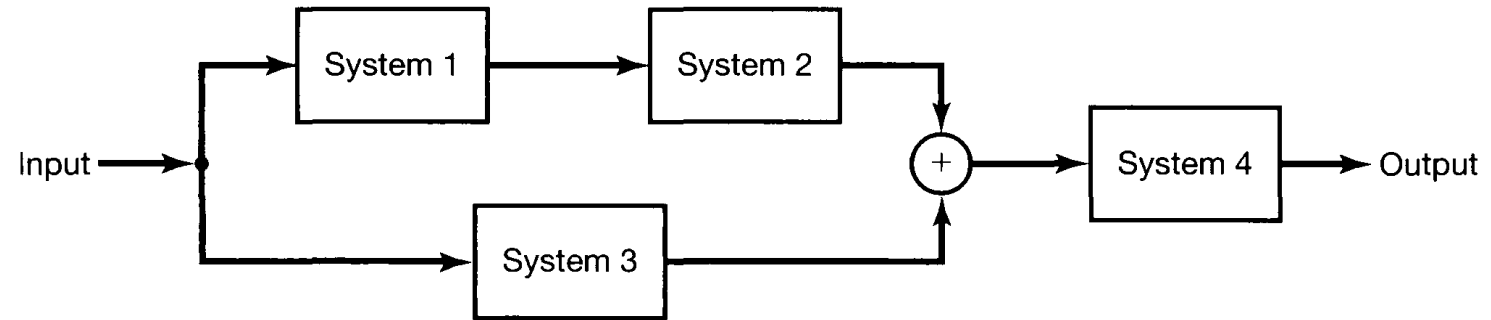


- Parallel Interconnection

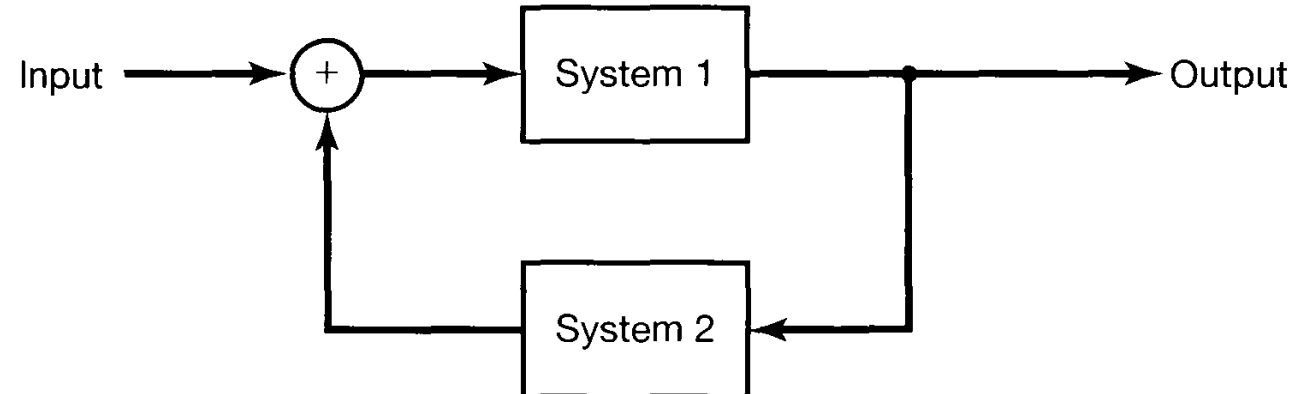


Interconnection of System:

- Series-Parallel Interconnection



- Feedback Interconnection





Time Domain analysis- LTIC Systems:

Convolution properties are similar in CT as we have seen in DT systems earlier

Commutative:

$$x[n] * h[n] = h[n] * x[n]$$

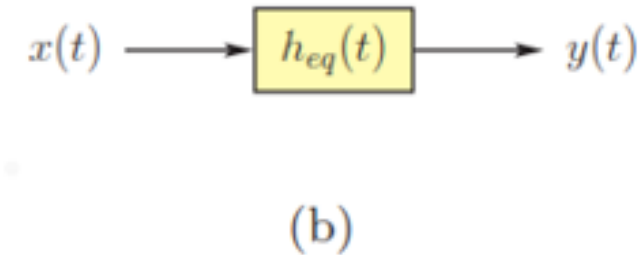
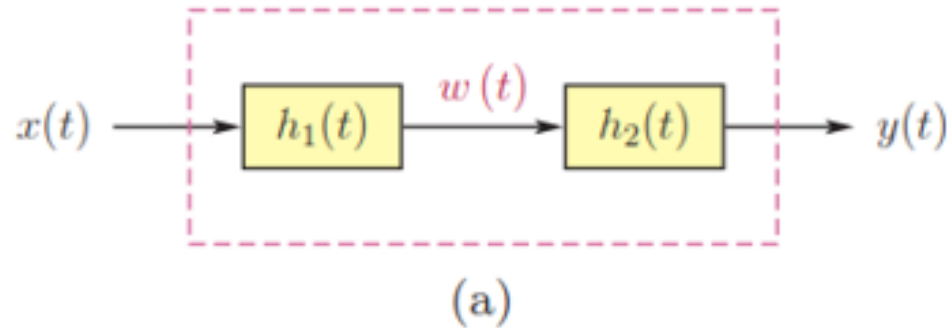
Associative:

$$\{x[n] * h_1[n]\} * h_2[n] = x[n] * \{h_1[n] * h_2[n]\}$$

Distributive:

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

Time Domain analysis- LTIC Systems:



$$w(t) = x(t) * h_1(t)$$

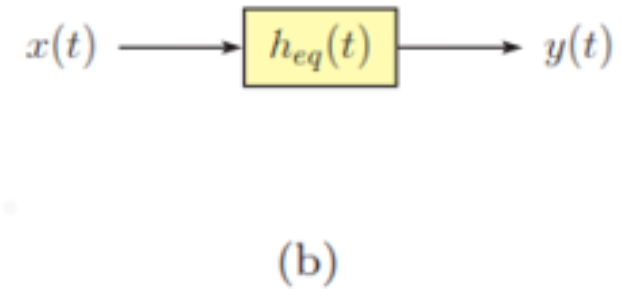
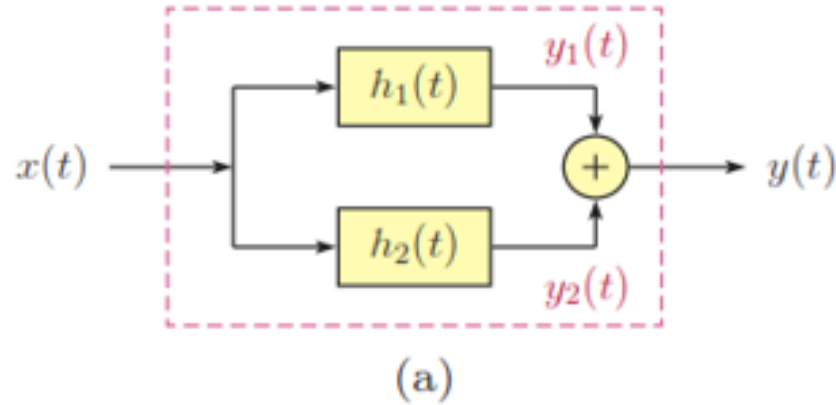
$$y(t) = w(t) * h_2(t)$$

$$y(t) = \{x(t) * h_1(t)\} * h_2(t)$$

$$y(t) = x(t) * \{h_1(t) * h_2(t)\}$$

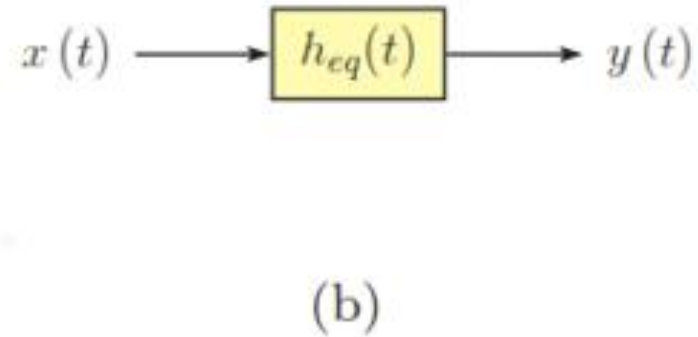
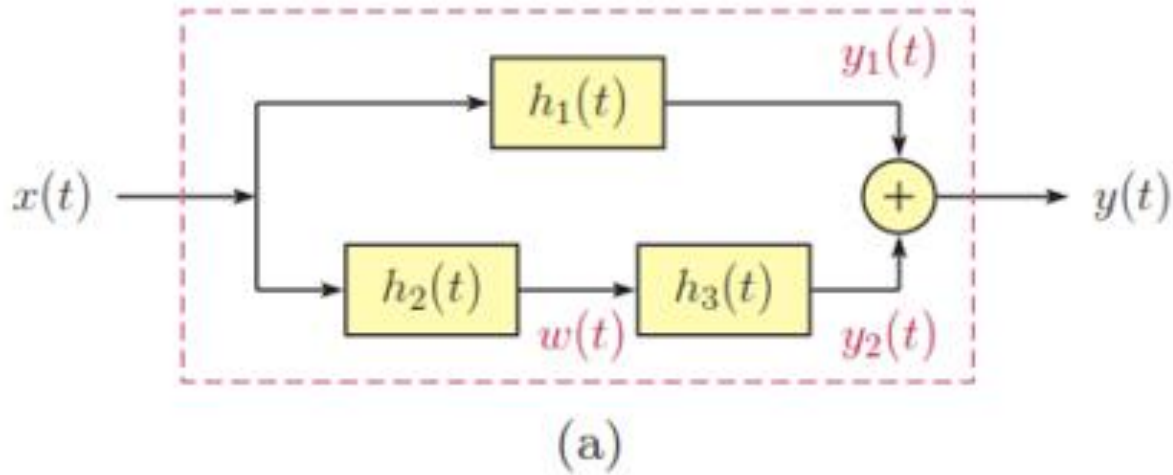
$$h_{eq}(t) = h_1(t) * h_2(t)$$

Time Domain analysis- LTIC Systems:



$$y(t) = x(t) * \{h_1(t) + h_2(t)\}$$
$$h_{eq}(t) = h_1(t) + h_2(t)$$

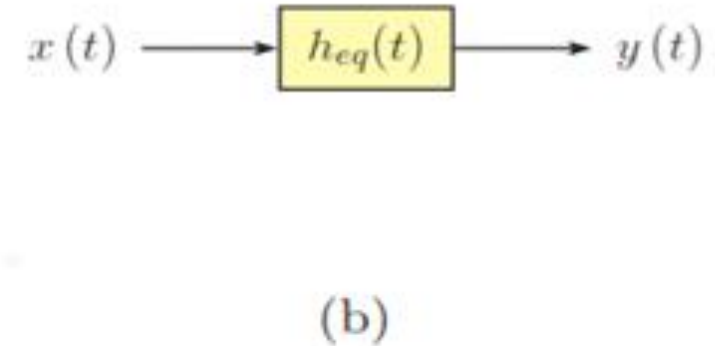
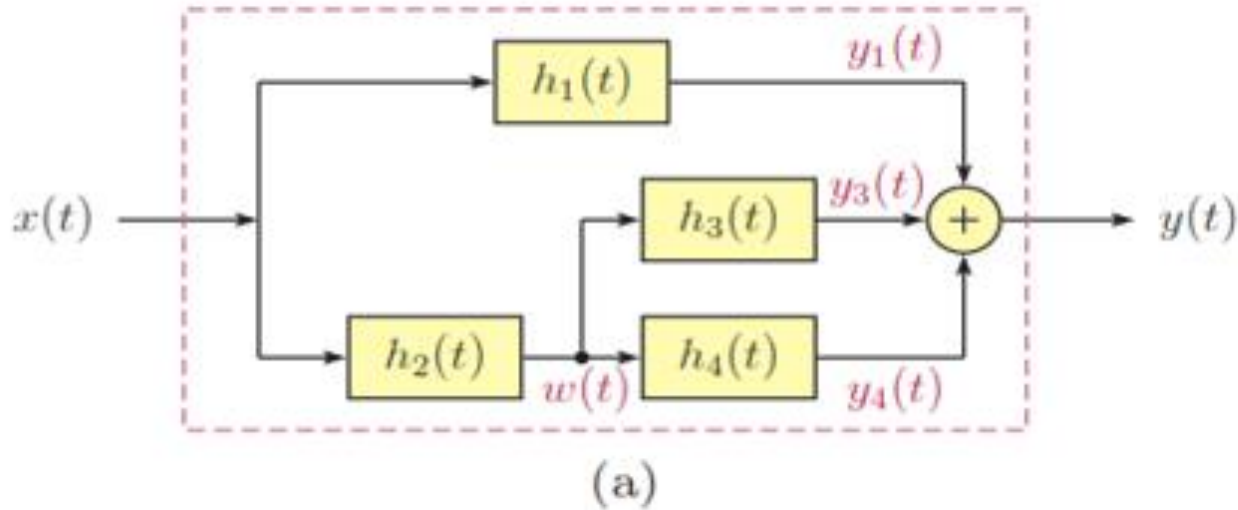
Time Domain analysis- LTIC Systems:



$$y(t) = x(t) * \{h_1(t) + h_2(t) * h_3(t)\}$$

$$h_{eq}(t) = h_1(t) + h_2(t) * h_3(t)$$

Time Domain analysis- LTIC Systems:



$$y(t) = x(t) * \{h_1(t) + h_2(t) * h_3(t) + h_2(t) * h_4(t)\}$$

$$h_{eq}(t) = h_1(t) + h_2(t) * h_3(t) + h_2(t) * h_4(t)$$



Time Domain analysis- LTIC Systems:

Properties of LTI Systems Using Impulse Response $h(t)$

Causality:

$$h(t) = 0 \quad \text{for } t < 0$$

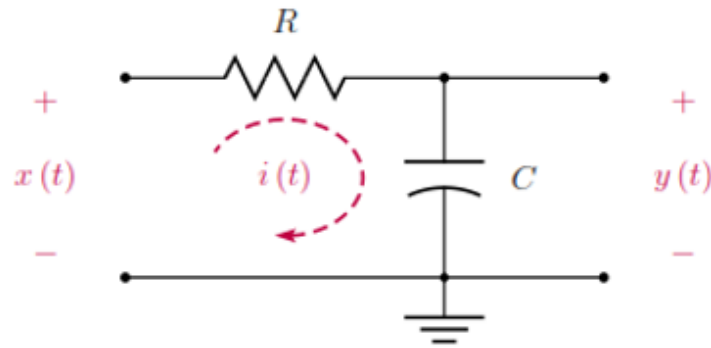
BIBO Stability:

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

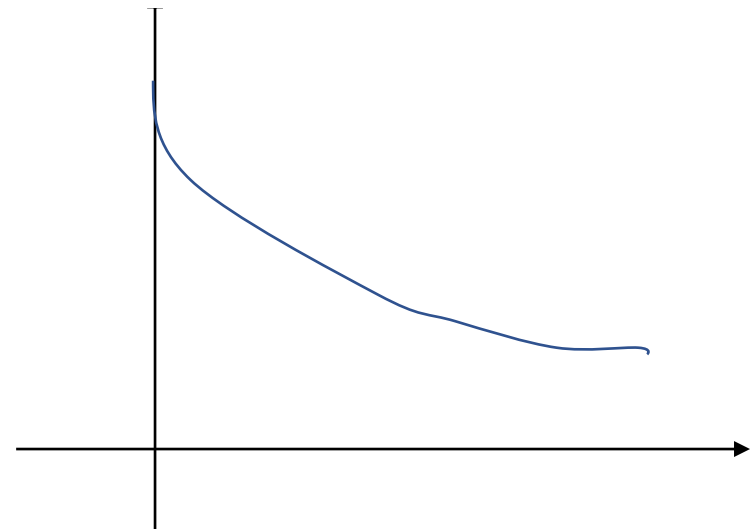
Time Domain analysis- LTIC Systems:

Properties of LTI Systems Using Impulse Response $h(t)$

Example:



$$h(t) = e^{-t}u(t)$$



$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

For $RC = 1$

$$h(t) = e^{-t}u(t)$$

Time Domain analysis- LTIC Systems:

Properties of LTI Systems Using Impulse Response $h(t)$

Example: $h(t) = e^{-t}u(t)$

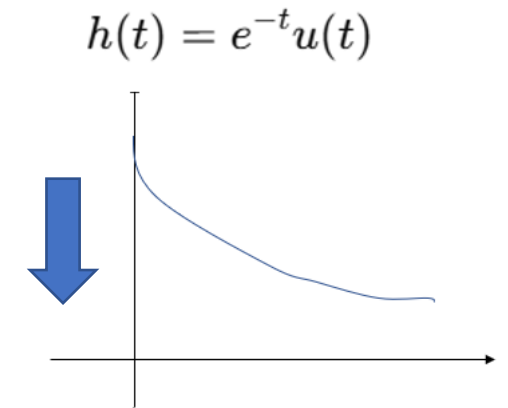
A system is causal if:

$$h(t) = 0 \quad \text{for } t < 0$$

Here, $u(t)$ (unit step) ensures:

$$h(t) = 0 \quad \text{for } t < 0$$

Therefore, given LTI system is Causal.





Time Domain analysis- LTIC Systems:

Properties of LTI Systems Using Impulse Response $h(t)$

Example: $h(t) = e^{-t}u(t)$

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

$$\begin{aligned} \int_{-\infty}^{\infty} |e^{-t}u(t)| dt &= \int_0^{\infty} e^{-t} dt \\ &= [-e^{-t}]_0^{\infty} = 1 \end{aligned}$$

Therefore, given LTI system is BIBO Stable.

End of session 01

End of Time Domain analysis- LTIC Systems.

Next: Frequency Domain analysis - LTIC Systems.



Learning outcome of Wk#12 session:

The time-domain analysis of linear, time-invariant, continuous-time (LTIC) systems.

Time Domain Analysis (Chapter 02)

- CT System modeling by Differential equations
- CT signal and system representations
- CT system impulse response $h(t)$
- CT Convolution- $y(t) = x(t) * h(t)$
- Properties of CT systems using $h(t)$:
 - Causality
 - BIBO Stability

Frequency Domain Analysis (Chapter 04)

- Eigenfunction of CT systems
- The Laplace Transform



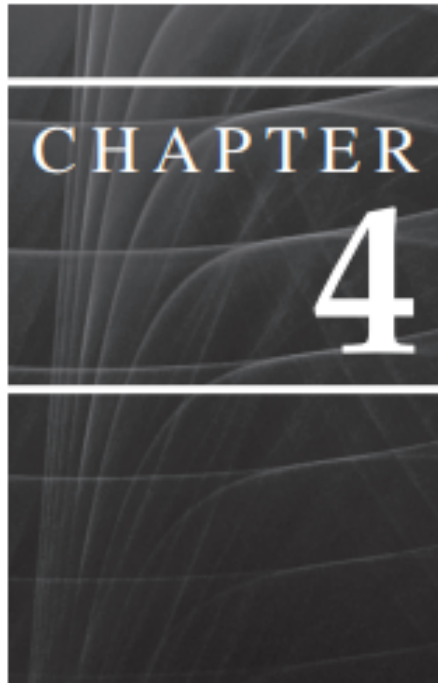
Analysis of LTIC Systems:

A quick comparison with LTI CT and DT systems:

- Notion of time changes from n to t , i.e., $y[n] \rightarrow y(t)$
- The summation operation $\sum x[n]$ changes to Integration $\int x(t)$
- The difference equations change to Differentiation equations to model the CT system
- Eigenfunction Z^n ($z = \gamma e^{j\Omega}$) changes to e^{st} ($s = \sigma + j\omega$)



Frequency Domain Analysis of LTIC Systems:



CONTINUOUS-TIME SYSTEM ANALYSIS USING THE LAPLACE TRANSFORM



Frequency Domain analysis- LTIC Systems:

For a CT signal $x(t)$, its Laplace Transform $X(s)$ is defined as

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$X(s)$ exist for only those values of S where the integral converges.

Here, $s = \sigma + j\omega$

$$x(t) \Longleftrightarrow X(s)$$

$$e^{st} = e^{(\sigma + j\omega)t}$$

$$e^{st} = e^{\sigma t} e^{j\omega t}$$

$e^{\sigma t} \rightarrow$ represents a growing/decaying factor

$e^{j\omega t} \rightarrow$ represents the frequency of the sinusoid



RECAP: Wk# 07 lecture Slide

Complex Exponential Signal Continuous Time (CT)

Useful Signals Models:



COMPLEX SINUSOID

$$e^{j\omega t}$$



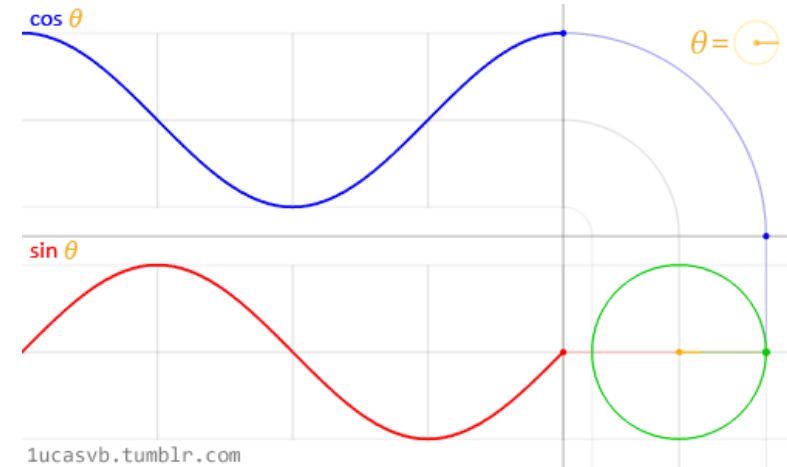
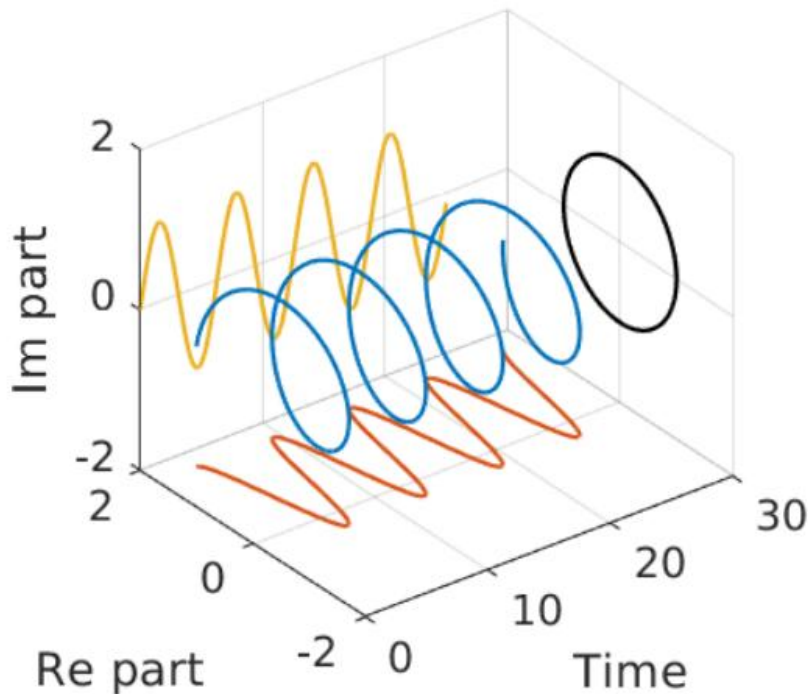
COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$

Useful Signals Models:

COMPLEX SINUSOID

$$e^{j\omega t}$$



- The complex sinusoid is a spiral, whose real and imaginary parts trace **cosine and sine**.
- The complex sinusoid traces a unit circle in the complex plane as its modulus is always 1.
- What if we allow the modulus to also grow or decrease with time? – need **Complex Exponential!**



Useful Signals Models:



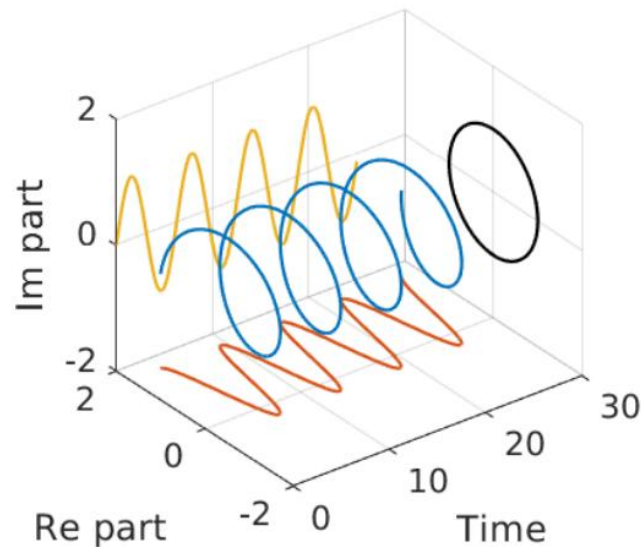
COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$

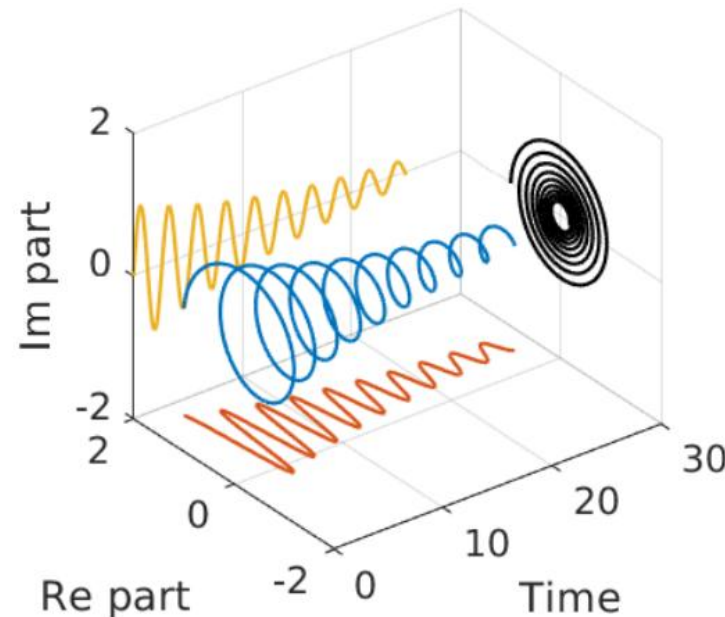
$$e^{(\sigma + j\omega)t} = e^{\sigma t} e^{j\omega t}$$

Complex Exponential is basically a complex Sinusoid with possibly decaying or growing amplitude.

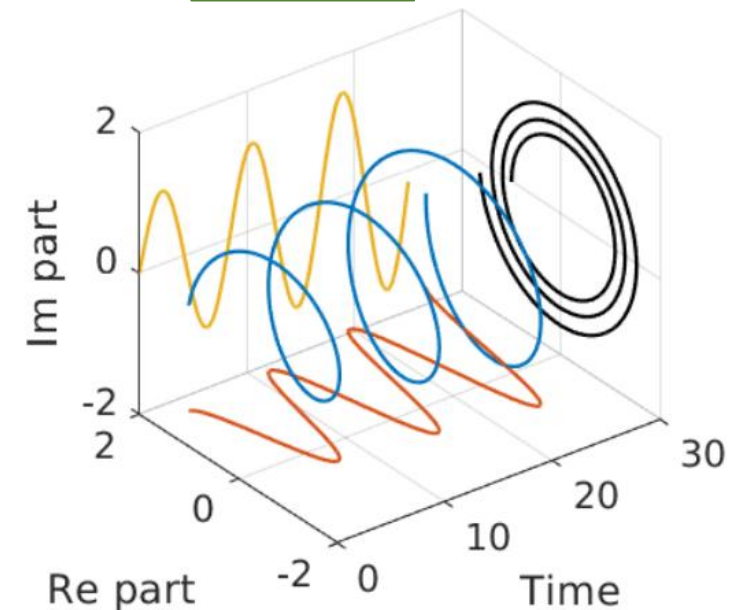
$$\sigma = 0$$



$$\sigma < 0$$



$$\sigma > 0$$





Useful Signals Models:

COMPLEX SINUSOID

$$e^{j\omega t}$$



Periodic with period $T_0 = \frac{2\pi}{\omega}$

COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$



Not periodic, unless $\sigma = 0$



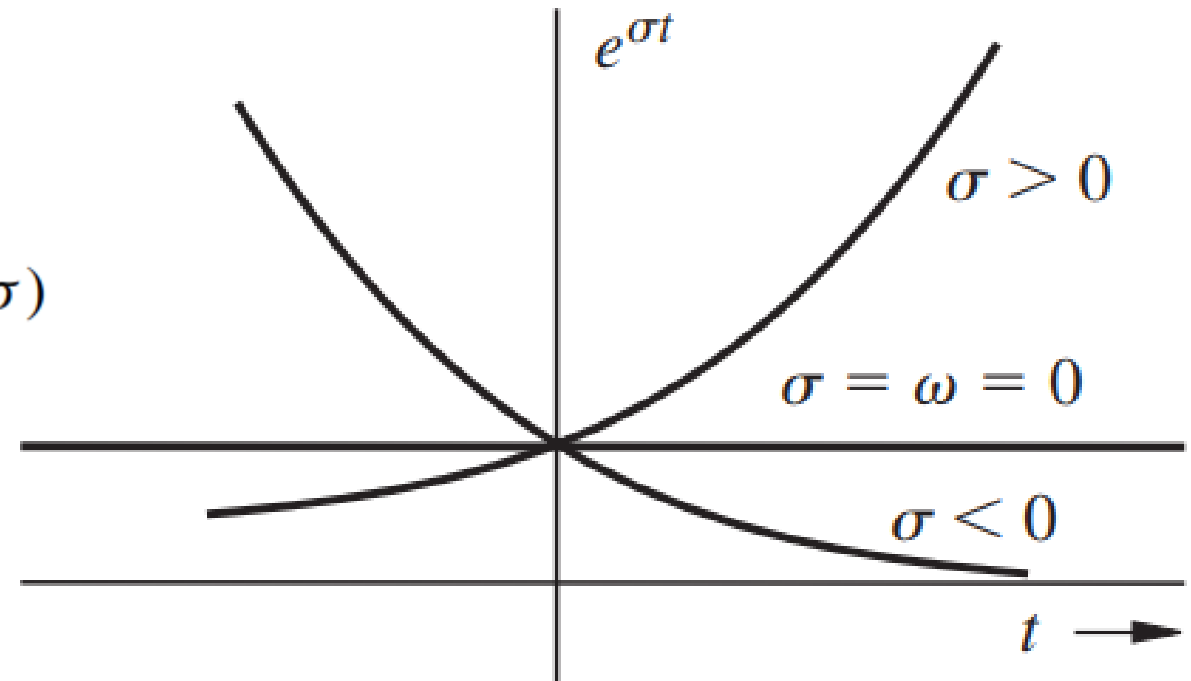
Useful Signals Models:

The exponential function e^{st}

The function e^{st} encompasses a large class of functions, for example

$$s = \sigma + j\omega$$

1. A constant $k = ke^{0t}$ ($s = 0$)
2. A monotonic exponential $e^{\sigma t}$ ($\omega = 0, s = \sigma$)

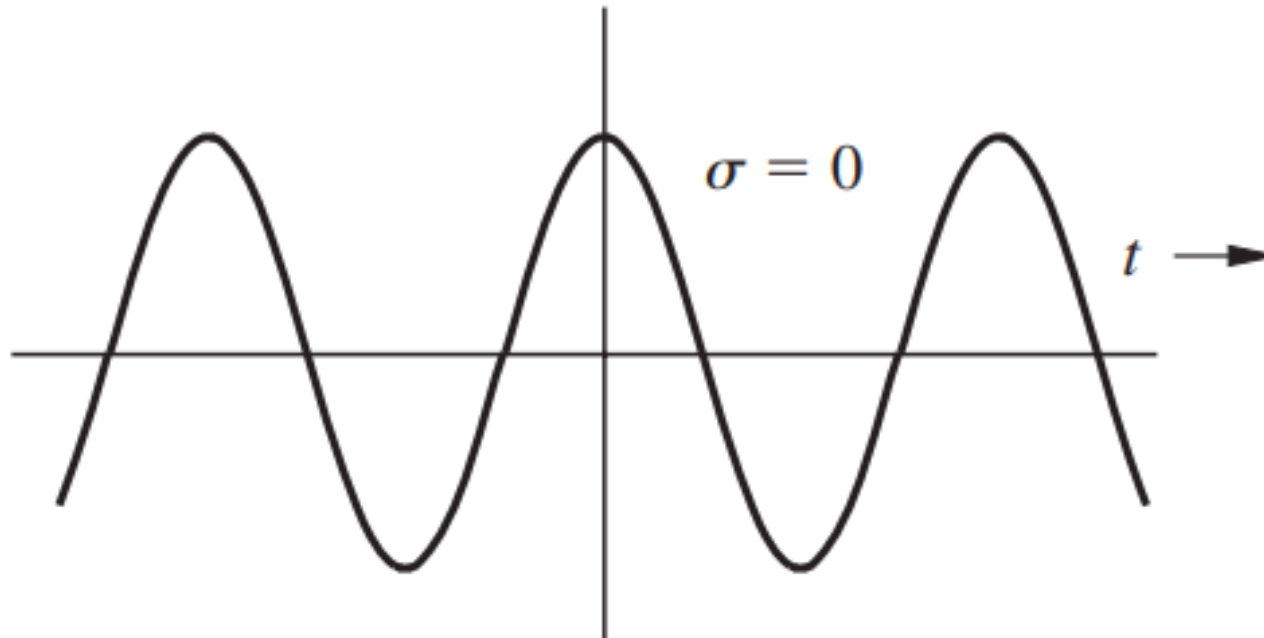


Useful Signals Models:

The exponential function e^{st}

$$s = \sigma + j\omega$$

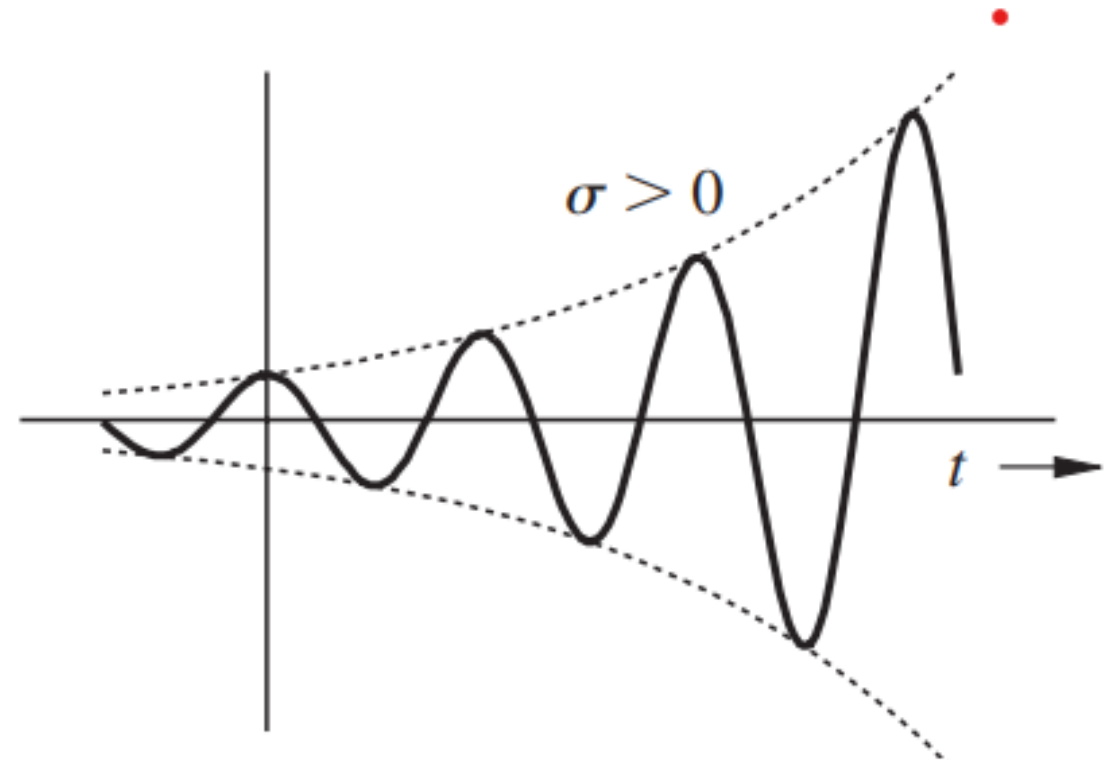
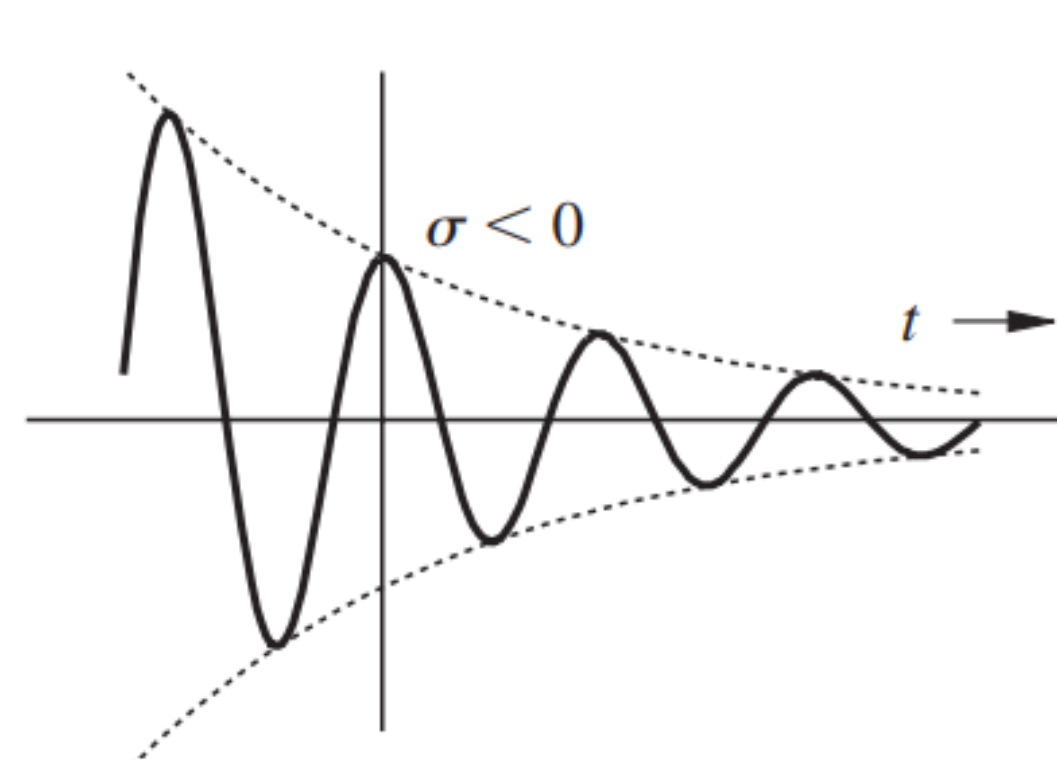
3. A sinusoid $\cos \omega t$ ($\sigma = 0, s = \pm j\omega$)



Useful Signals Models:

The exponential function e^{st}

4. An exponentially varying sinusoid $e^{\sigma t} \cos \omega t$ ($s = \sigma \pm j\omega$)





Useful Signals Models:

The exponential function e^{st}

The complex frequency “ s ” is represented on a complex frequency plane (s plane). The s plane is separated into two parts:

- The left half-plane (LHP) corresponds to exponentially decaying signals
- The right half-plane (RHP) corresponds to exponentially growing signals.

The imaginary axis separates the two regions and corresponds to signals of constant amplitude.

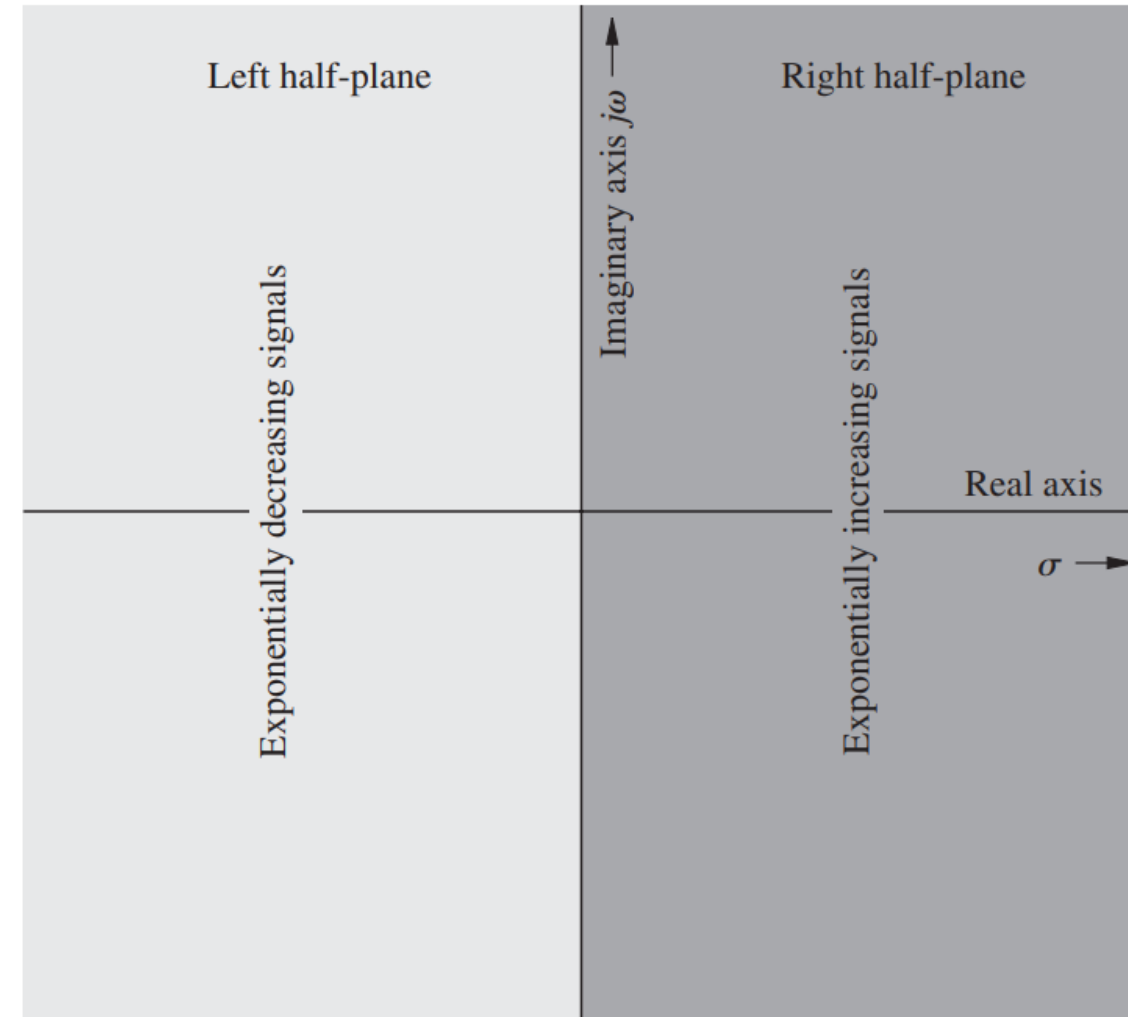


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

An exponentially growing sinusoid $e^{2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(2+j5)t}$ and $e^{(2-j5)t}$ with **complex frequencies $2 + j5$ and $2 - j5$** , respectively, which lie in the RHP.

Working: $e^{\sigma t} \cos \omega t = \frac{1}{2}(e^{st} + e^{s^*t})$

$$s = \sigma + j\omega$$

For given, $e^{2t} \cos 5t \Rightarrow$ here $\sigma = 2$ and $\omega = 5$
therefore, $s = 2 + j5$ and $s^* = 2 - j5$

Cos (ωt) – a real signal (not imaginary) must have a conjugate complex frequency pair

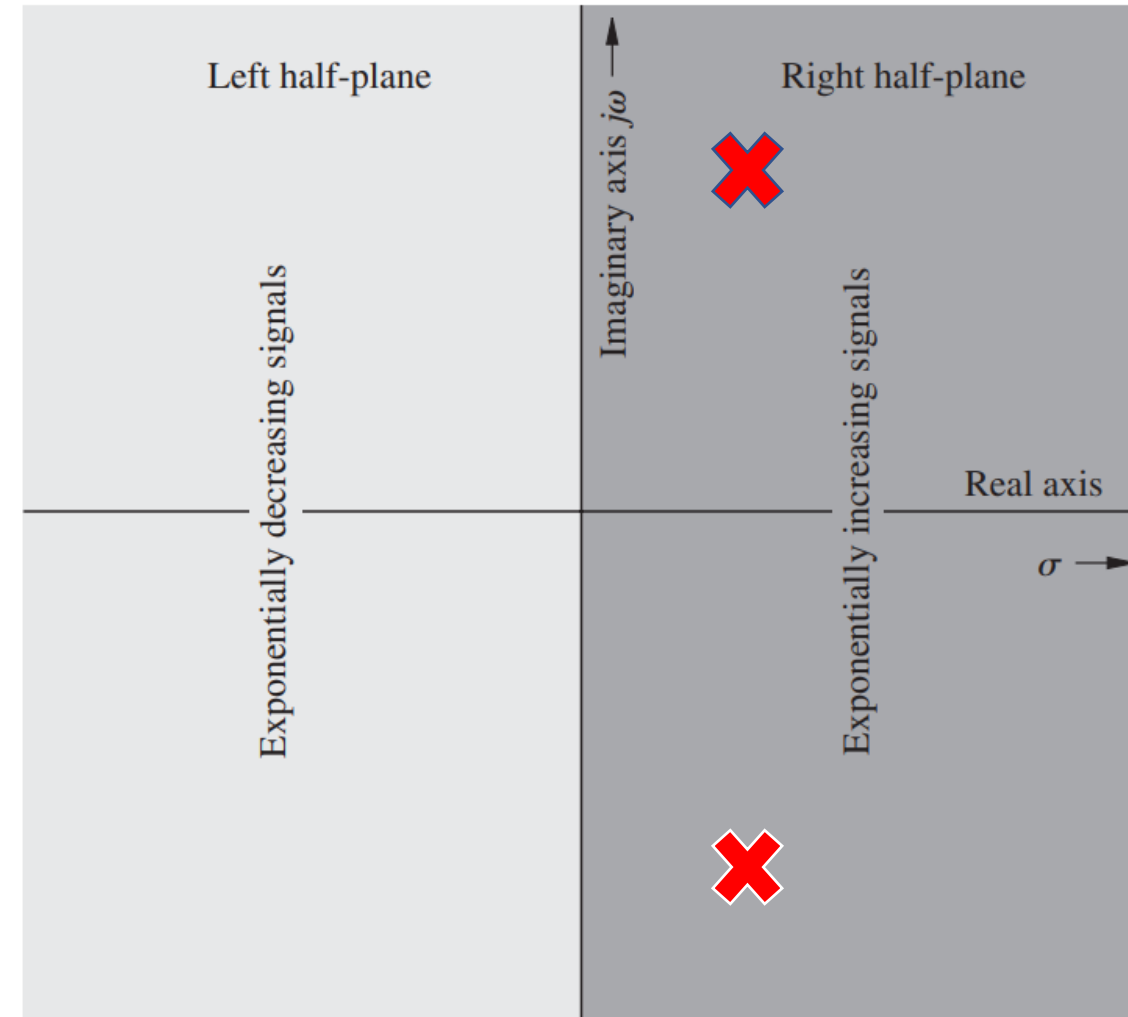


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

An exponentially growing sinusoid $e^{2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(2+j5)t}$ and $e^{(2-j5)t}$ with complex frequencies $2 + j5$ and $2 - j5$, respectively, which lie in the RHP.

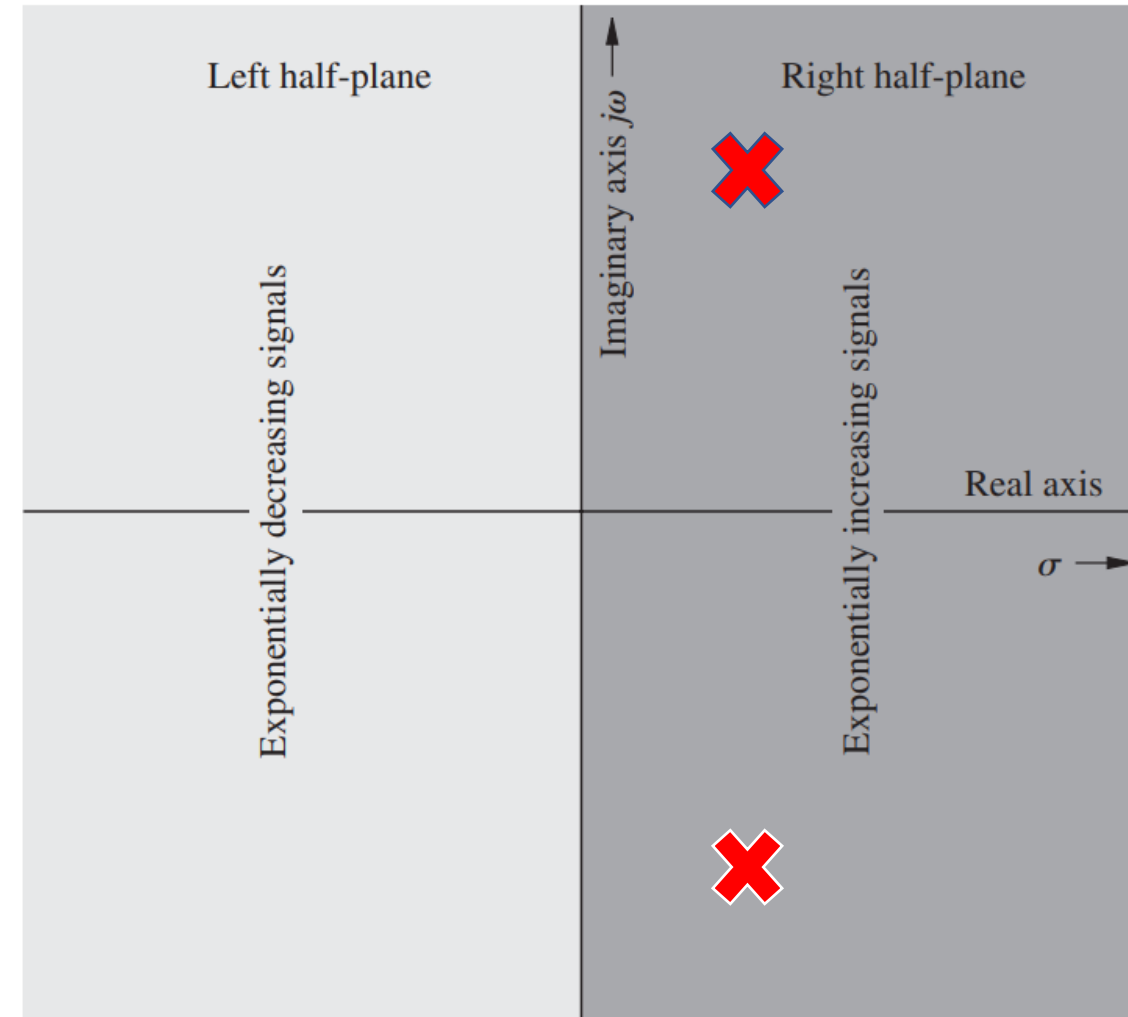
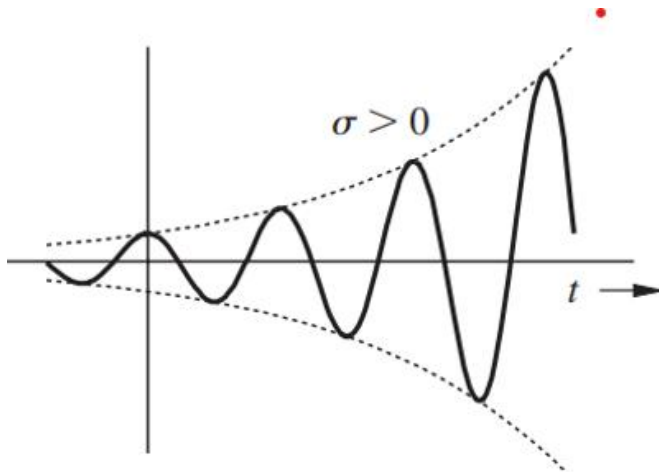


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

An exponentially decaying sinusoid $e^{-2t} \cos 5t$, for example, can be expressed as a linear combination of exponentials $e^{(-2+j5)t}$ and $e^{(-2-j5)t}$ with complex frequencies $2 + j5$ and $2 - j5$, respectively, which lie in the LHP.

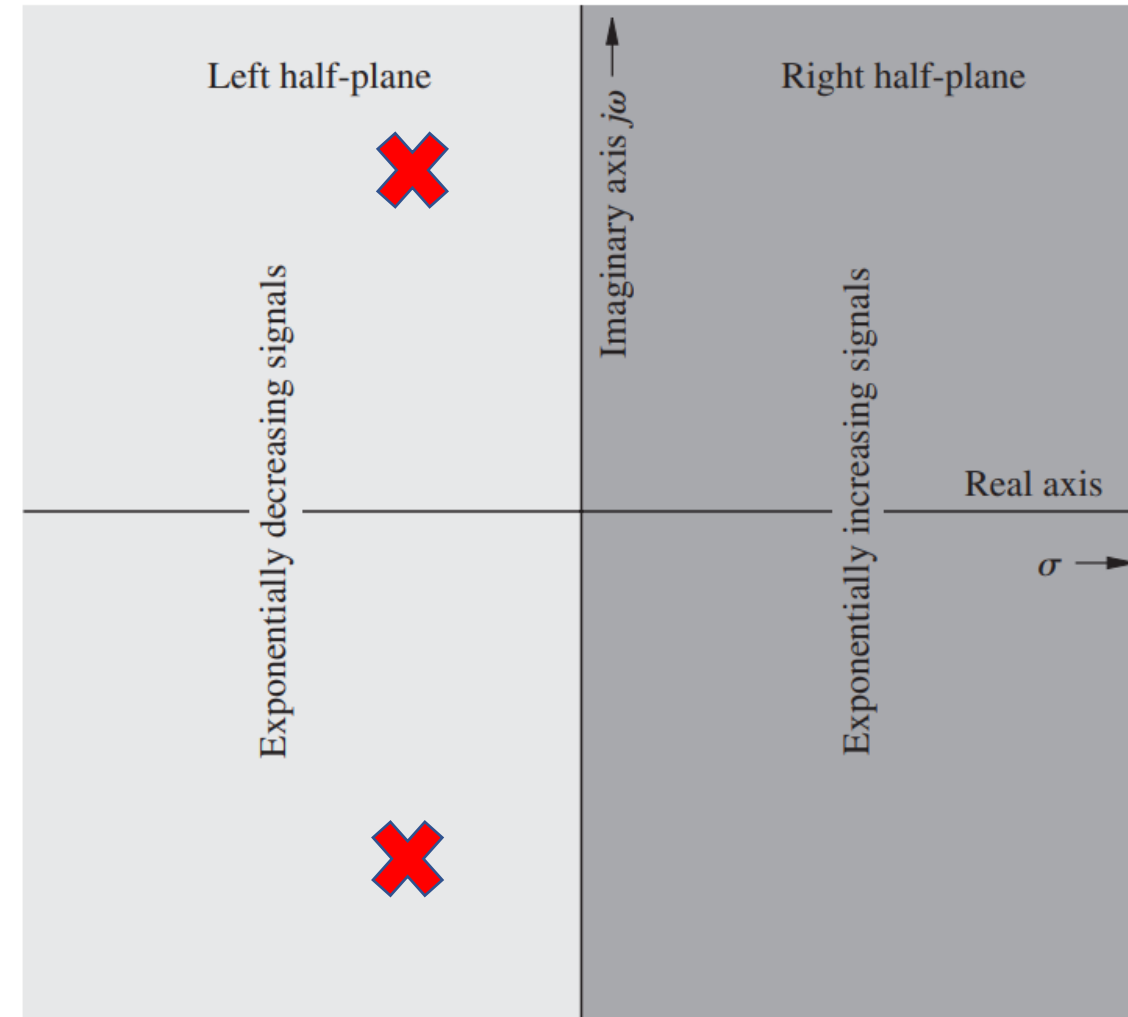
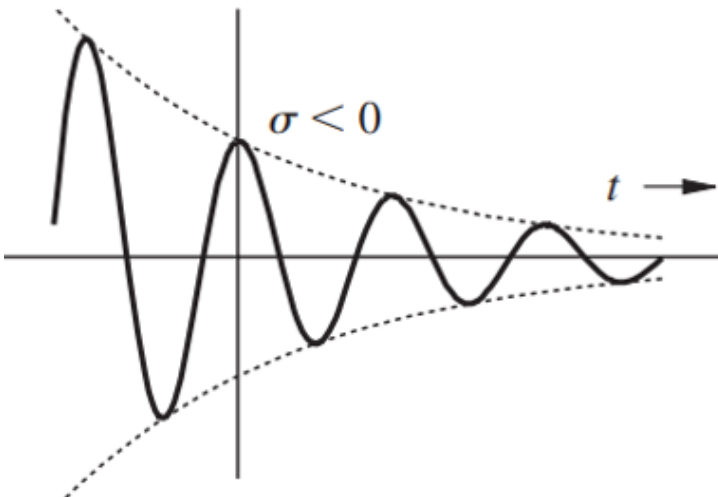


Figure 1.22 Complex frequency plane.

Useful Signals Models:

The exponential function e^{st}

A constant-amplitude sinusoid $\cos 5t$ can be expressed as a linear combination of exponentials e^{j5t} and e^{-j5t} with **complex frequencies $\pm j5$** , which lie on the imaginary axis.

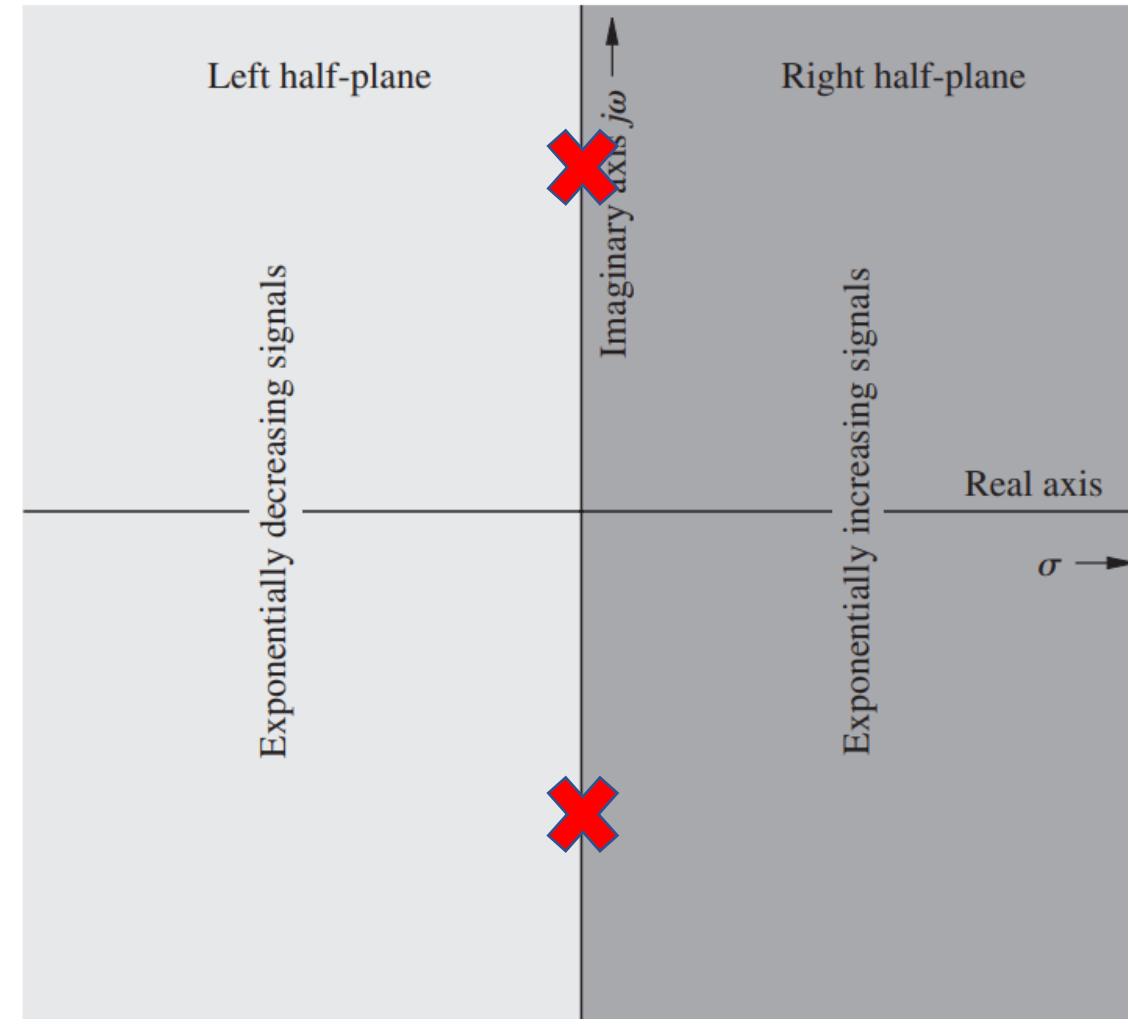
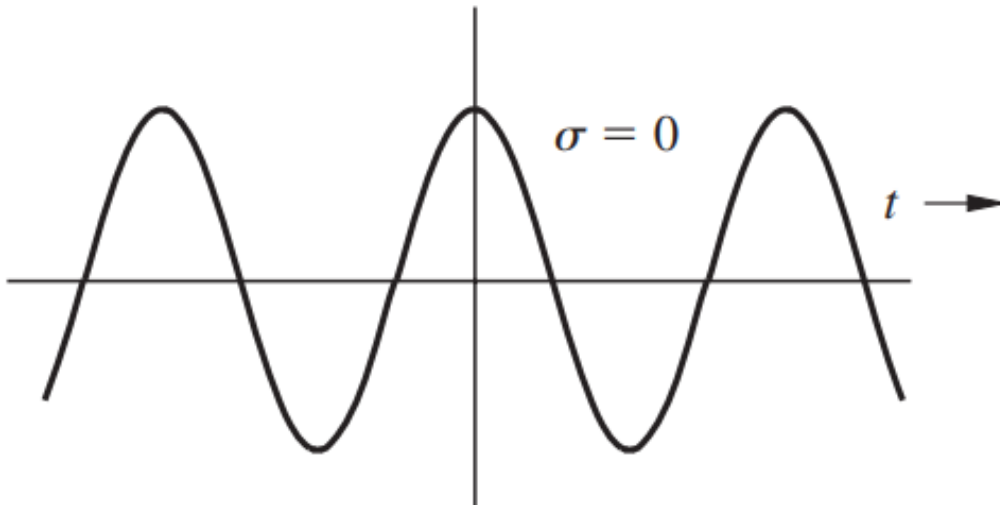


Figure 1.22 Complex frequency plane.

The S plane:

Stable Signal/Systems:

Left-hand side of the S plane, exponential decay.

Unstable Signal/Systems:

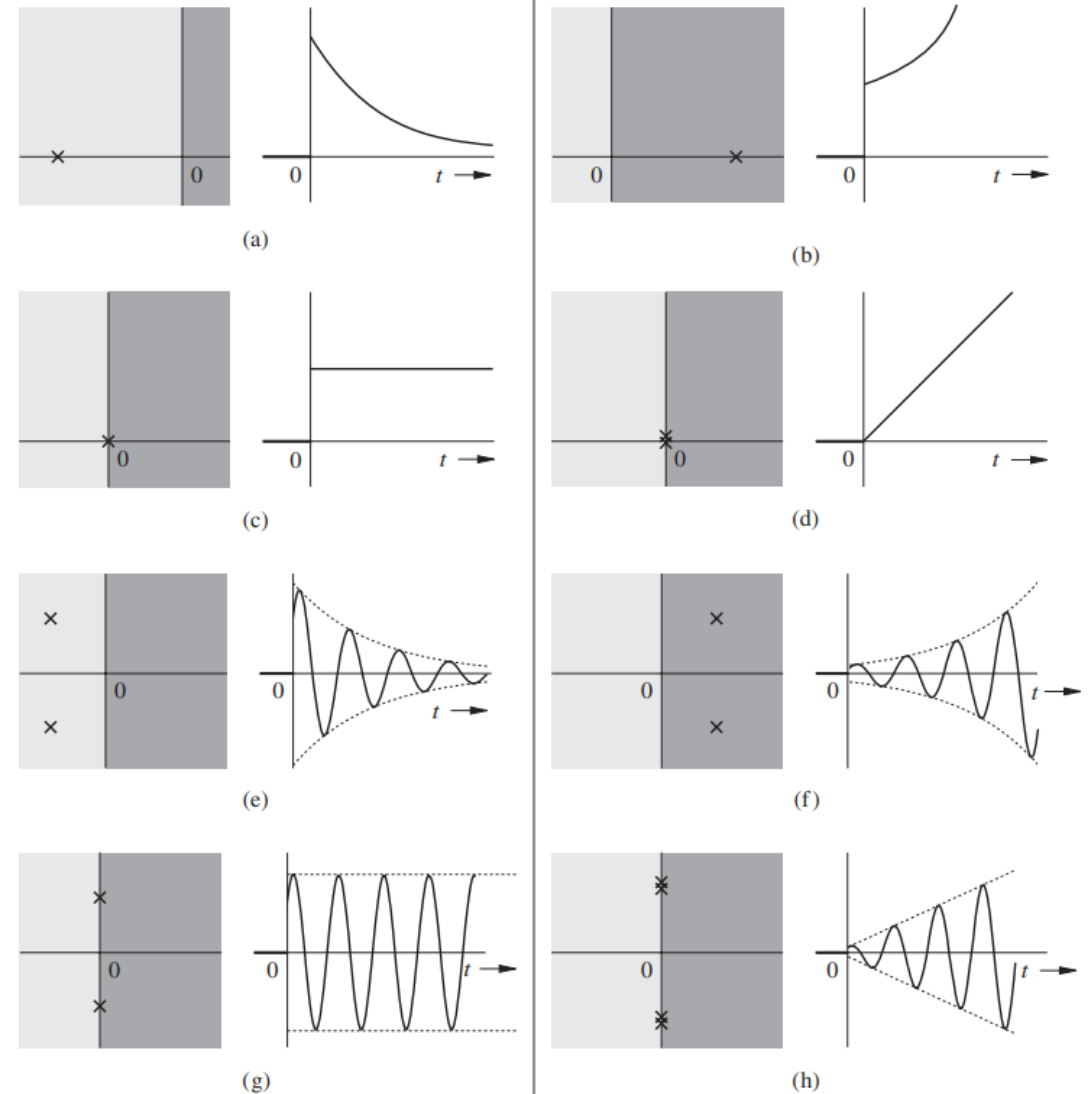
Right-hand side of the S plane, exponential growth

Marginally Stable Signal/Systems:

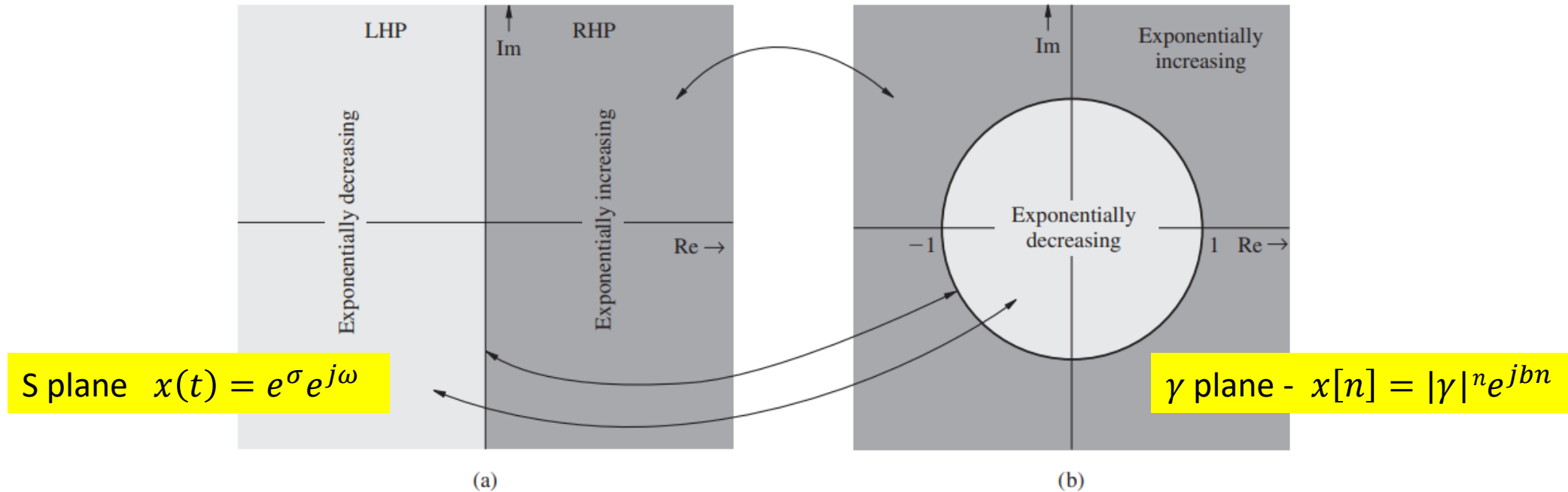
$\sigma = 0$ (Y-axis), oscillation (no decay).

One more condition:

Repeated poles



The S-plane vs Z-Plane:



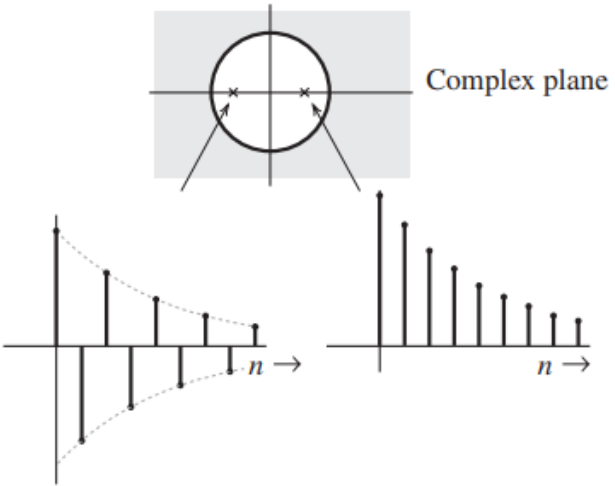
For $\sigma < 0$: exponentially decreasing

$\sigma > 0$: exponentially increasing

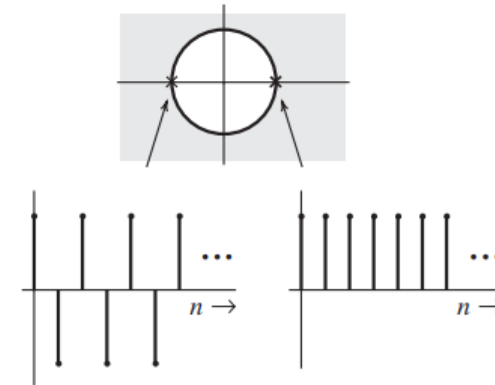
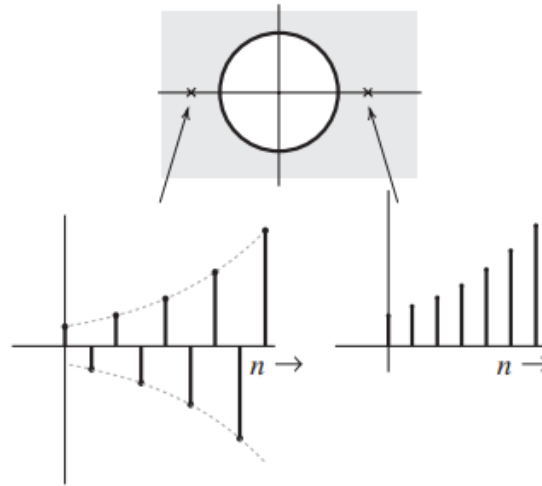
$\sigma = 0$: constant magnitude

- For $|\gamma| > 1$, $\gamma^n \rightarrow \infty$ as $n \rightarrow \infty$
- For $|\gamma| < 1$, $\gamma^n \rightarrow 0$ as $n \rightarrow \infty$
- For $|\gamma| = 1$, $\gamma^n = 1$ for all n

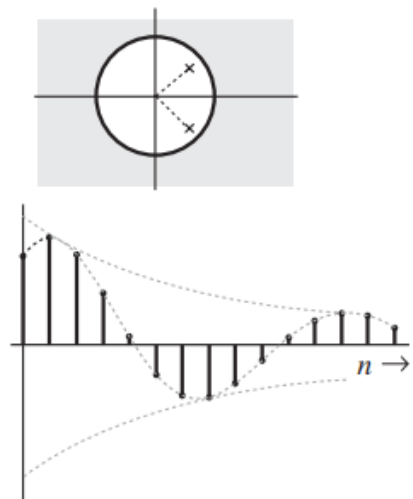
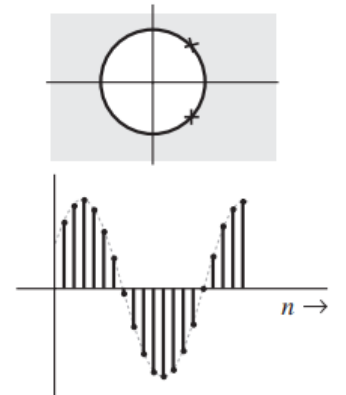
The S-plane vs Z-Plane:



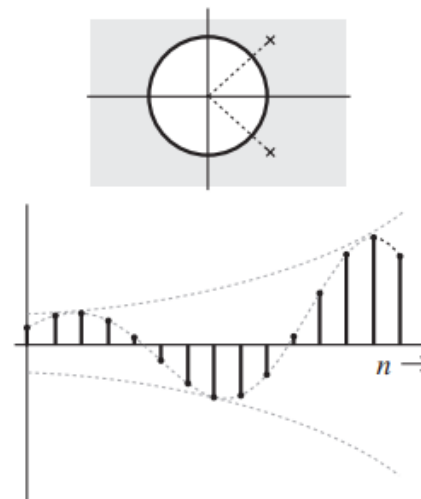
(a)



(c)



(b)



Back to Laplace Transform



Frequency Domain analysis- LTIC Systems:

Bilateral Laplace Transform.

$$X(s) = \int_{-\infty}^{+\infty} x(t) e^{-st} dt$$

Unilateral Laplace Transform.

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt$$

For Causal Signals

Frequency Domain analysis- LTIC Systems:

For a signal $x(t) = e^{-at}u(t)$, find the Laplace transform $X(s)$ and its ROC.

By definition,

$$X(s) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st} dt$$

Because $u(t) = 0$ for $t < 0$ and $u(t) = 1$ for $t \geq 0$,

$$X(s) = \int_0^{\infty} e^{-at}e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty}$$

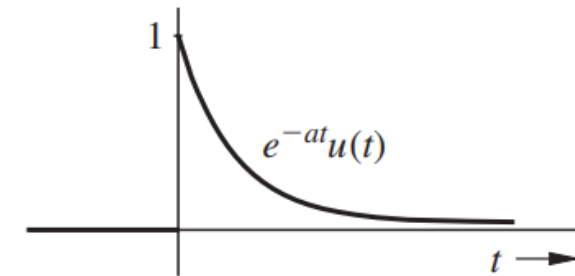
$$= \frac{1}{-(s+a)} [(e^{-(s+a)\cdot\infty}) - (e^{-(s+a)\cdot 0})]$$

$$\lim_{t \rightarrow \infty} e^{-(s+a)t} = \begin{cases} 0 & \text{Re}(s+a) > 0 \\ \infty & \text{Re}(s+a) < 0 \end{cases}$$

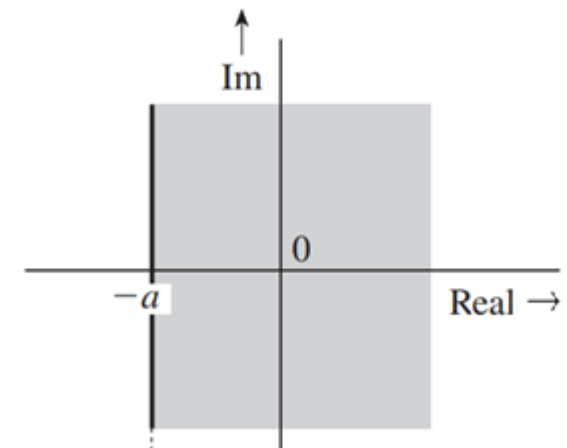
$$X(s) = \frac{1}{s+a} \quad \text{Re}(s+a) > 0$$

$$e^{-at}u(t) \iff \frac{1}{s+a} \quad \text{Re } s > -a$$

Signal $x(t)$



Region of convergence



Quiz-III

Thanks!